

Advanced Natural Language Processing

Multimodal Models

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2026-2027



Maastricht University

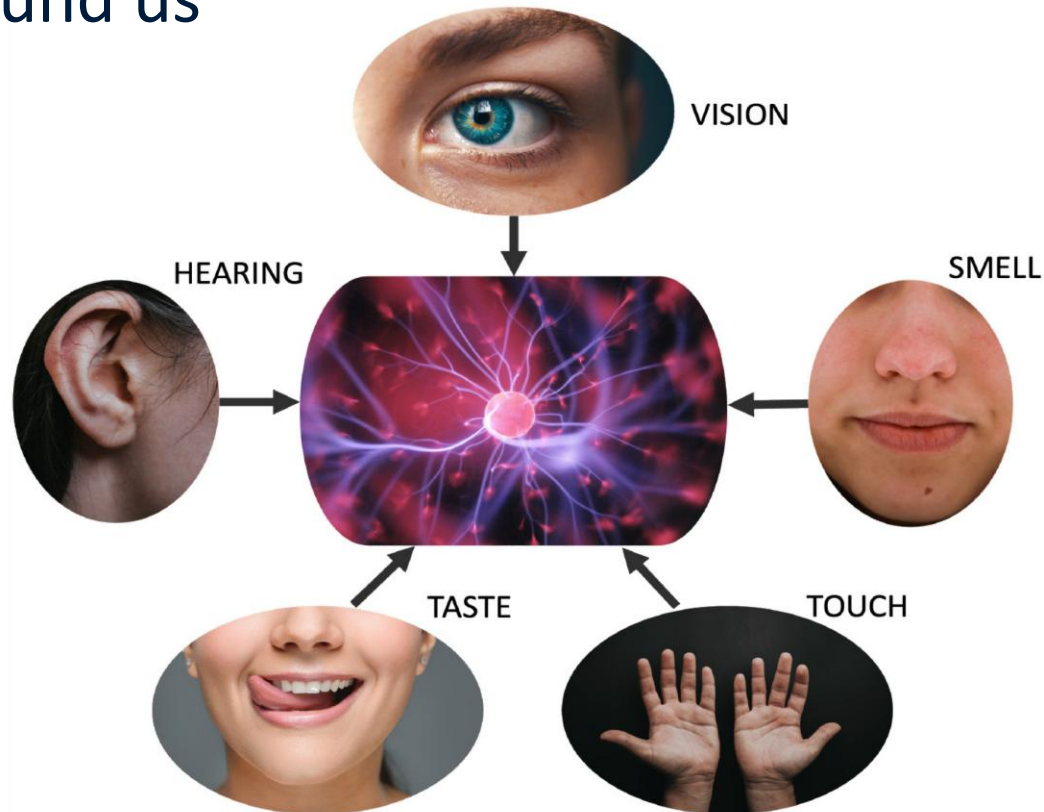
Multimodality

= Having more than one modality

- What is a modality?
 - Text, speech, information extracted by text and audio/speech analysis
 - Images, video, information extracted by computer vision
 - Sensor data, logs, structured data
 - Control of a robot, machinery, circuitry
 - Any physical or non-physical entity (car, emotion,...)

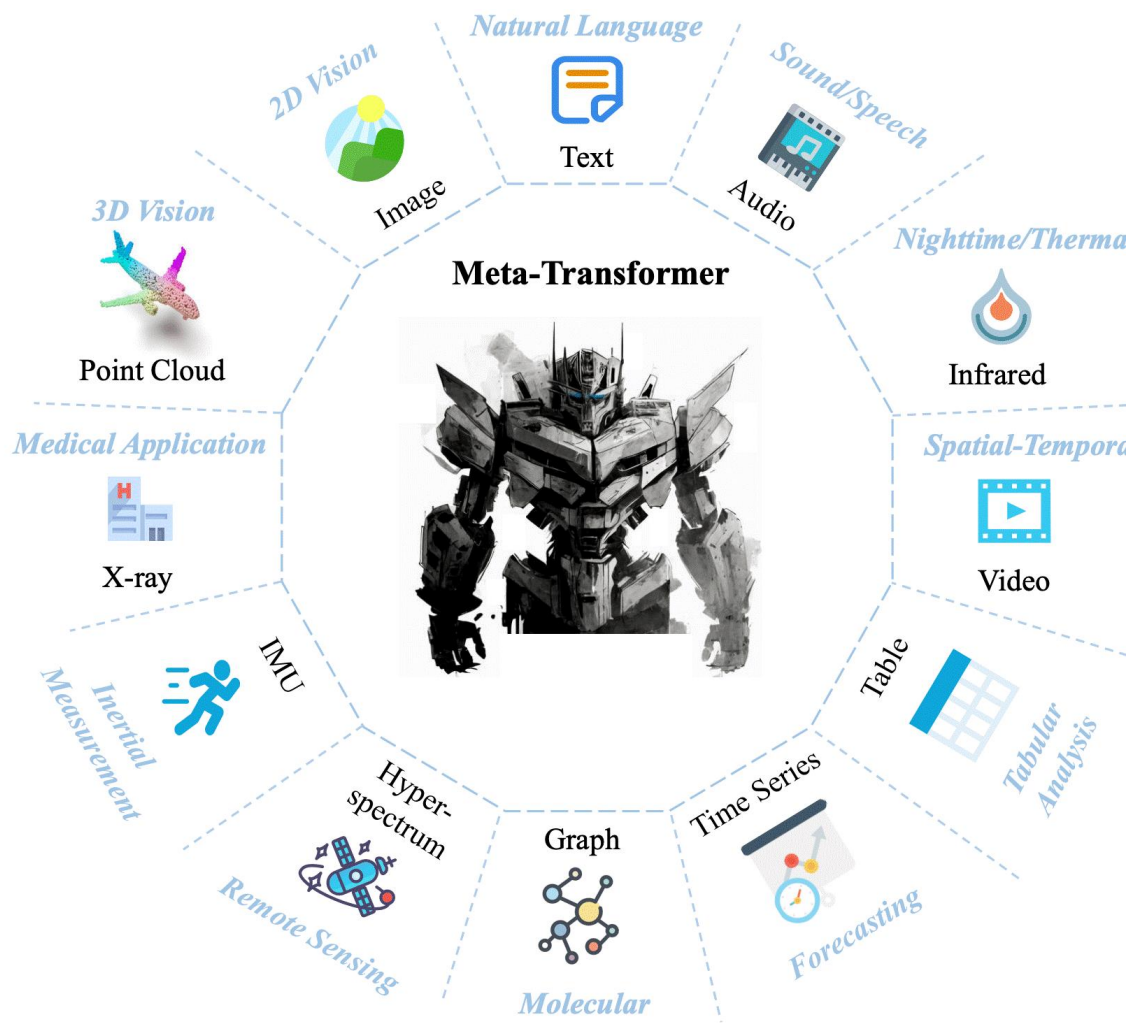
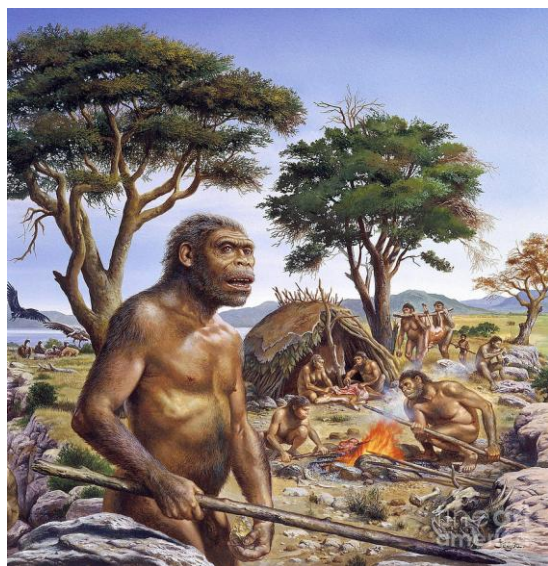
We as humans are also multimodal

Our senses provide different interpretations to the world around us



When text is not enough

Humans are multimodal agents – unimodal interaction is rare



Multimodal history

Visionary history of multimodal processing and alignment that never really took off



THE SEMANTIC WEB

A new form of Web content
that is meaningful to computers
will unleash a revolution of new possibilities

by
TIM BERNERS-LEE,
JAMES HENDLER and
ORA LASSILA

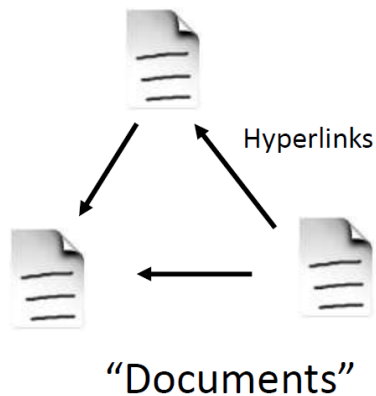
2001



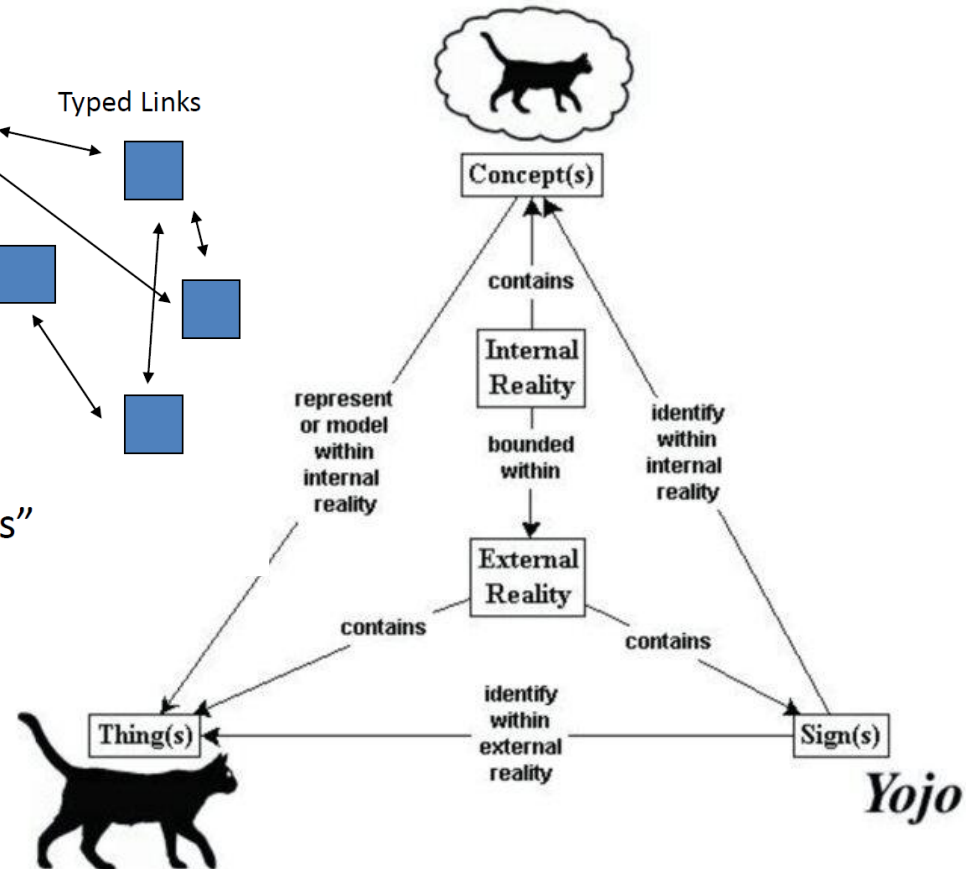
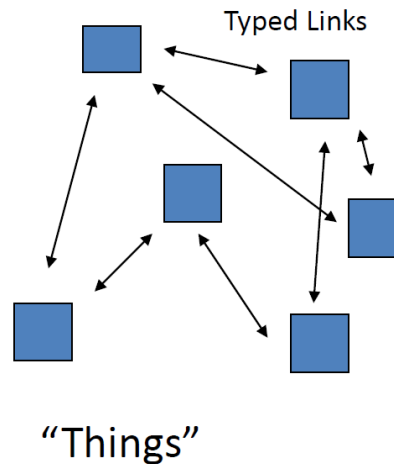
History: semantic information retrieval

- Semantic Web, “Web 3.0”, 1960’s, 2000’s

Web of Documents

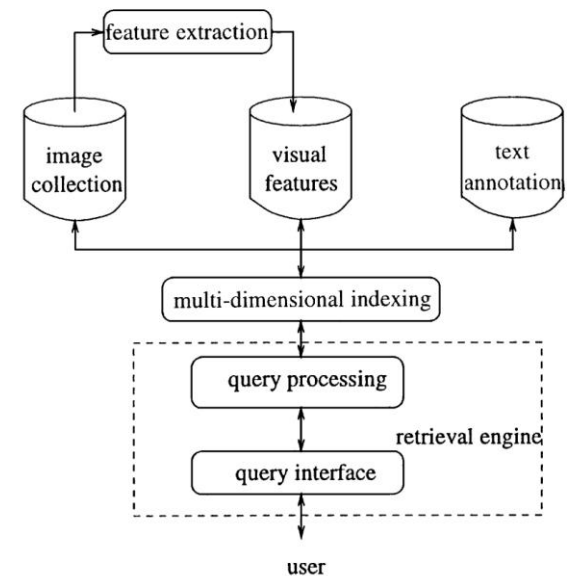


Web of Data



Multimedia Information Retrieval (MIR)

- Framework for text/image/audio/etc content representation and query
 - IBM's QBIC (1995)
 - Google's VisualRank (2008)



Multimedia Information Retrieval (MIR)

- MPEG-7 Standard for MIR (2000's)

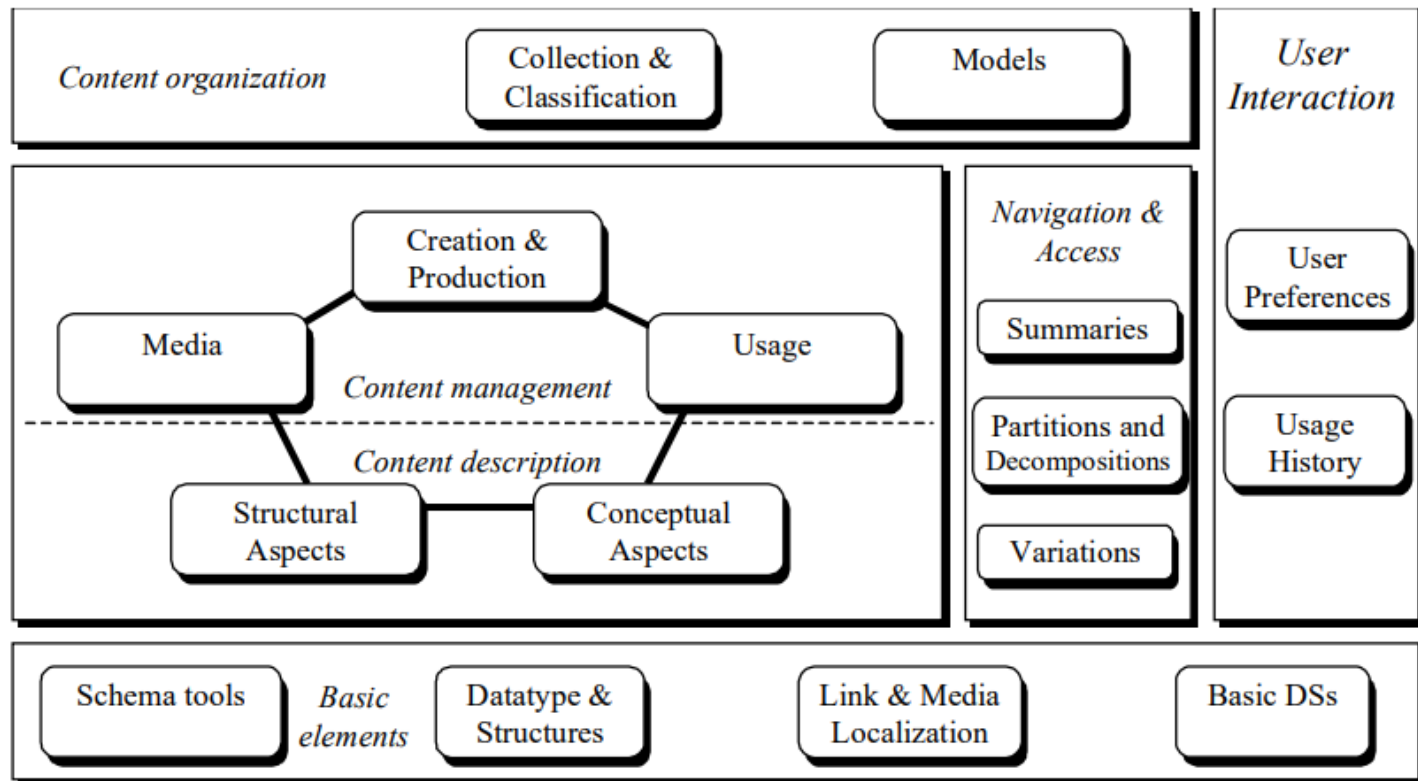
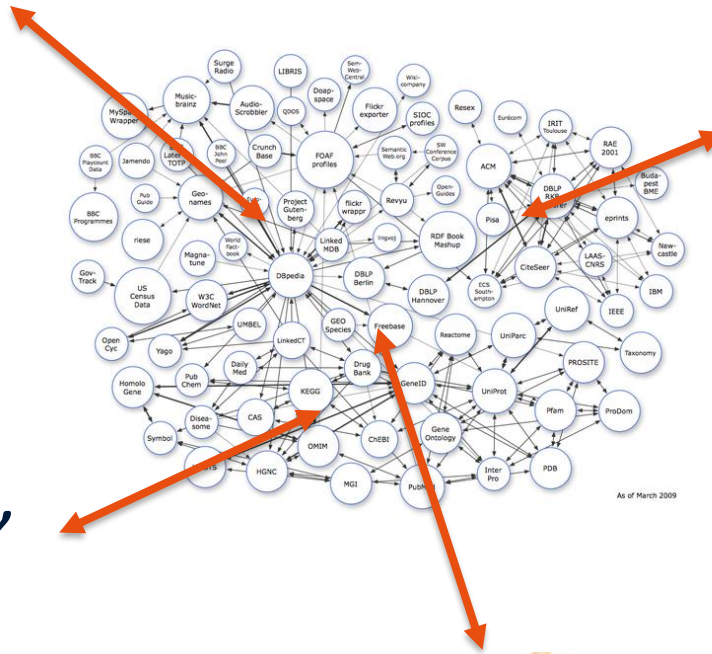


Figure 4: Overview of MPEG-7 Multimedia Description Schemes

Symbolic semantic models

“Yojo (cat) and I went to the park today and she had great time there 😊 “



“meow!”

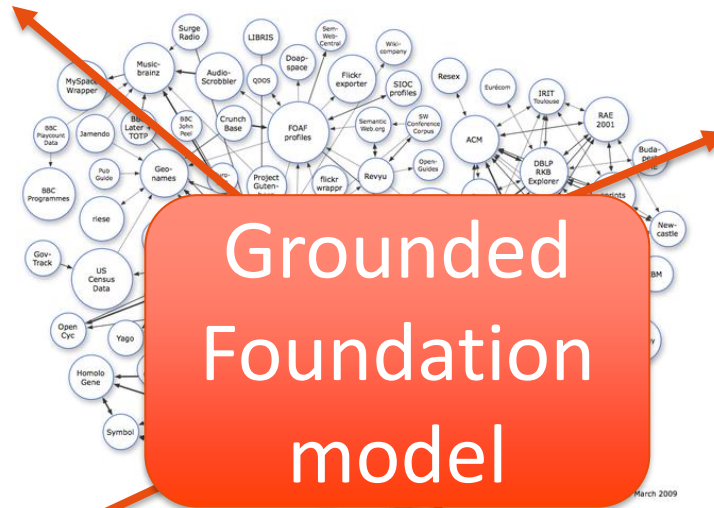


Fully explainable, but,

- A huge work to build/maintain a symbolic knowledge model
- Difficult to model uncertainty, ambiguities

Neuro-symbolic semantic models: *grounding*

“Yojo (cat) and I went to the park today and she had great time there 😊 “



“meow!”

Combines both advantages and some of the disadvantages of both

Neural semantic models

“Yojo (cat) and I went to the park today and she had great time there 😊 “

Multimodal
Foundation
model



“meow!”

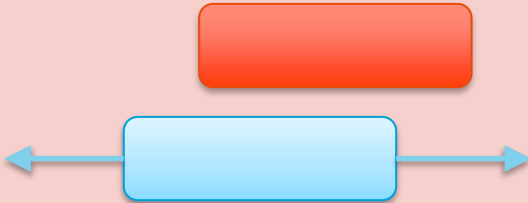
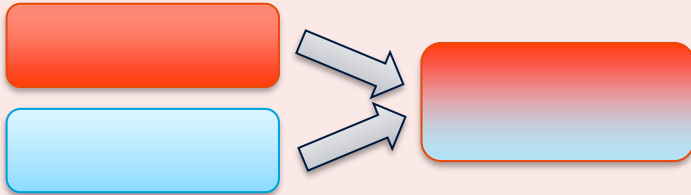
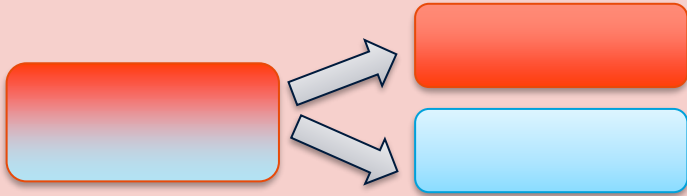


As good as our data is, but,

- Not explainable/debuggable
- Requires a lot of data
- Difficult to maintain

Basic concepts

Multimodal processing tasks

Transform		Text-to-speech, text-to-image, image-to-caption, speech-to-robot control
Alignment		Speech/text alignment, autodubbing, sensing/robot movement, cross-modal attention
Fusion, generation		Audio-visual speech recognition, Visual Question Answering, Editing
Fission, understanding		Audio-visual diarization, segmentation, speech-to-intent/emotion

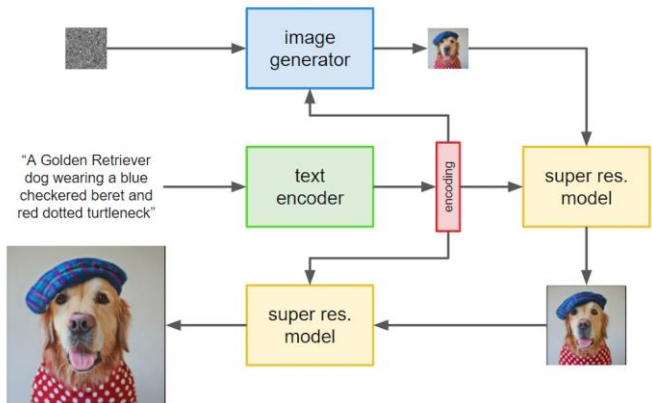
Terminology

- Equivalent terms:
 - VLM – Vision Language Model
 - MLLM – Multimodal Large Language Model
 - MMLM – Multi-Modal Language Model
- Applications:
 - VQA – Visual Question Answering
 - Document Understanding
 - Multimodal Summarization
 - VLA – Visual, Language, Action model (robotics)

Text-to-image

Many different types of image generation models exist.

GANs vs Diffusion



Flux-Pro



Flux-Dev



Flux-Schnell



SD3



Dall-E 3



Ideogram



SDXL



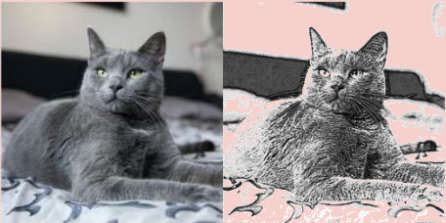
Multimodal Image Editing - Local

Object Manipulation	Attribute Manipulation	Spatial Transformation
"... with a glasses" 	"...is standing" 	Object Translation 
"Remove red flowers" 	"A fluffy (↑) doll" 	Object Scaling 
"... basketball football ..." 	"...round square..." 	Local Distortion 
Replacement by Exemplar Image 	Appearance Transfer 	

Multimodal Image Editing - Global

Style Change

"... in penciled style"



"Turn it to Van Gogh style"



Style Transfer



Image Translation

"A room"



"A toy"



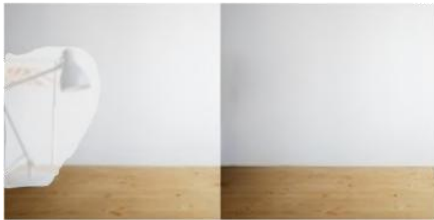
"A man"



Inpainting and Outpainting

- Inpainting: region within the image:

" "



"A parrot"



"A vase "



Inpainting with Exemplar Image



- Outpainting: region outside of the image:

"17th century kitchen background"



(Composed) Image Retrieval

Reference image



Relative caption

has two children
instead of cats



Target Images



is on a track and
has the front
wheel in the air

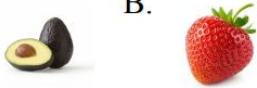


Visual Question Answering



User: Which one of the following can be found in the first image?

A. B.



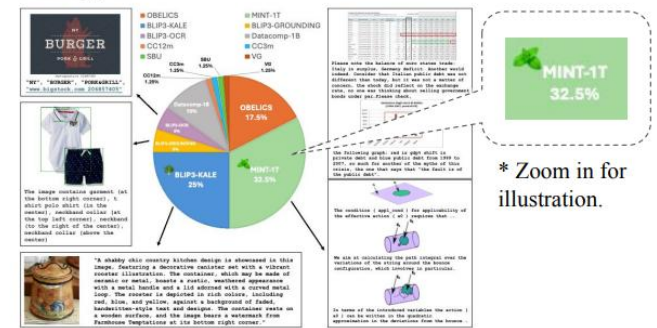
xGen-MM: B



User: Where does the object in image 1 appear in image 2?

xGen-MM: The object in image 1, a screwdriver, appears in image 2 on the left side, near the top left corner.

As indicated in Figure 3 in BLIP-3, we pre-train on an ensemble of diverse multimodal datasets with the indicated sampling ratios.



User: How much of BLIP-3's pre-training data is from MINT-1T?

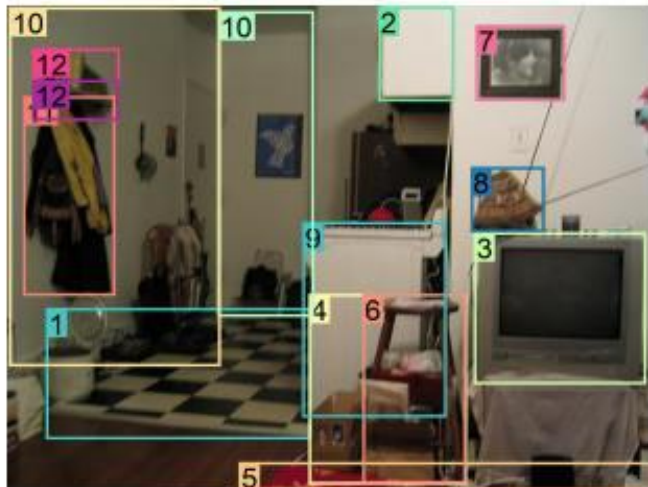
xGen-MM: MINT-1T contributes 32.5% of BLIP-3's pre-training data.

Multimodal Object Alignment

Prompt: Locate the phrases in the caption: {caption}



The image shows a group of five cartoon monsters. On the left side, there is a brown monster¹ with horns and a big smile on its face. Next to it, there are two smaller monsters², one black and one green. The black monster³ has two large horns on its head and is standing in the center of the group. The green monster⁴ on the right side is a green monster with big eyes and a long antennae. It is standing on its hind legs with its arms stretched out to the sides. In the middle of the image, there appears to be a small blue monster⁵ with a round head and two antennae on its back. The background is light beige with small green circles scattered around.

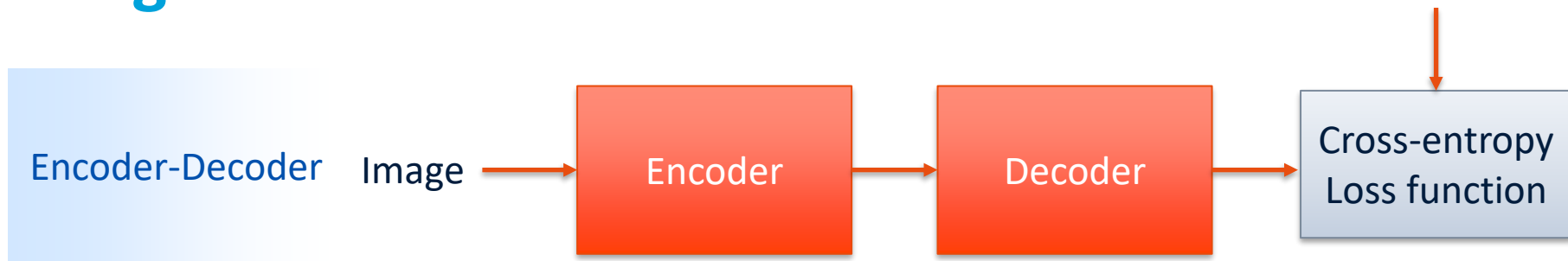


The image shows a cluttered room with a black and white checkered floor¹. On the right side of the image, there is a small white cabinet² with a television³ on top of it. Next to the cabinet, there are several items⁴ scattered on the floor, including a red blanket⁵, a wooden stool⁶, and a pile of trash. On top of the cabinet is a picture frame⁷ and a hat⁸. In the center of the room is a white refrigerator⁹ with a few items on top. The walls¹⁰ are painted white and there are a few clothes¹¹ hanging on a rack¹² on the left wall. The room appears to be in disarray, with some items strewn about and others scattered around.

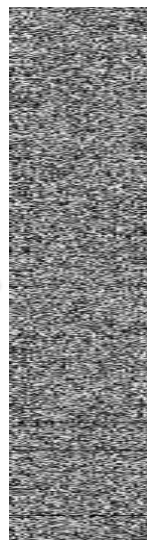
Architectures and model optimization



Image-to-Text: encoder-decoder formulations



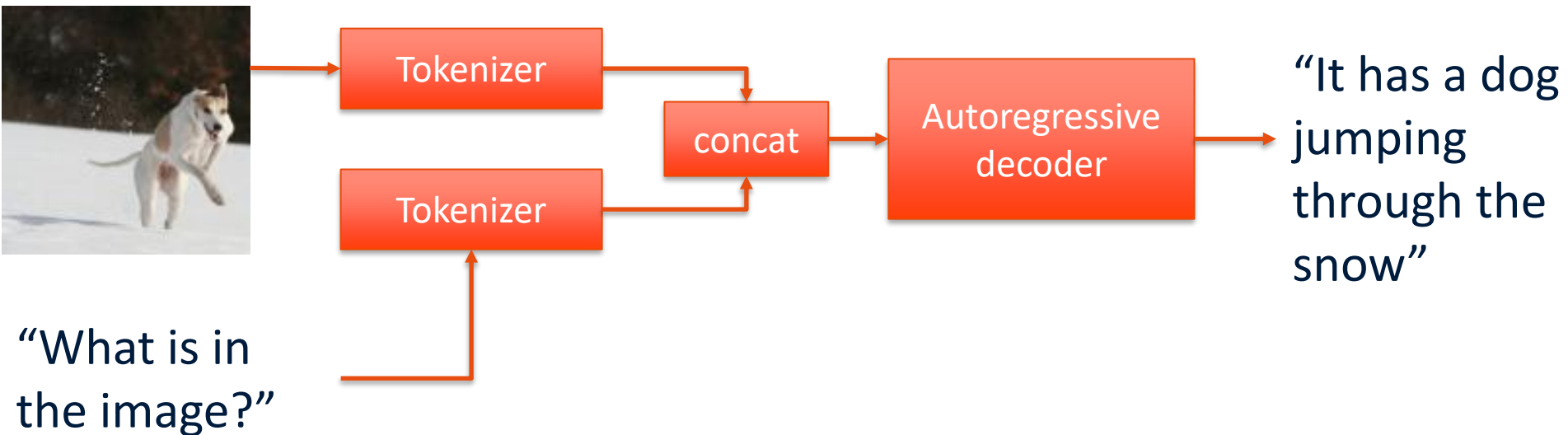
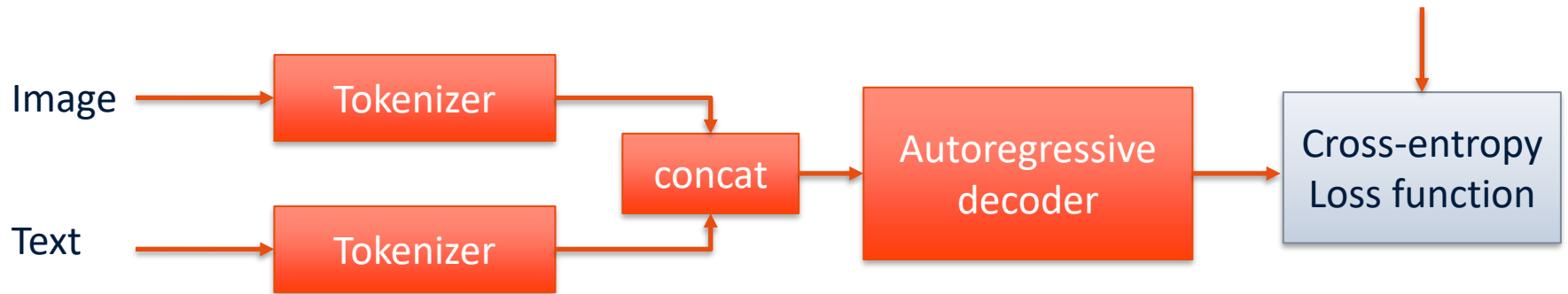
$3 \times 128 \times 128 = 49152$



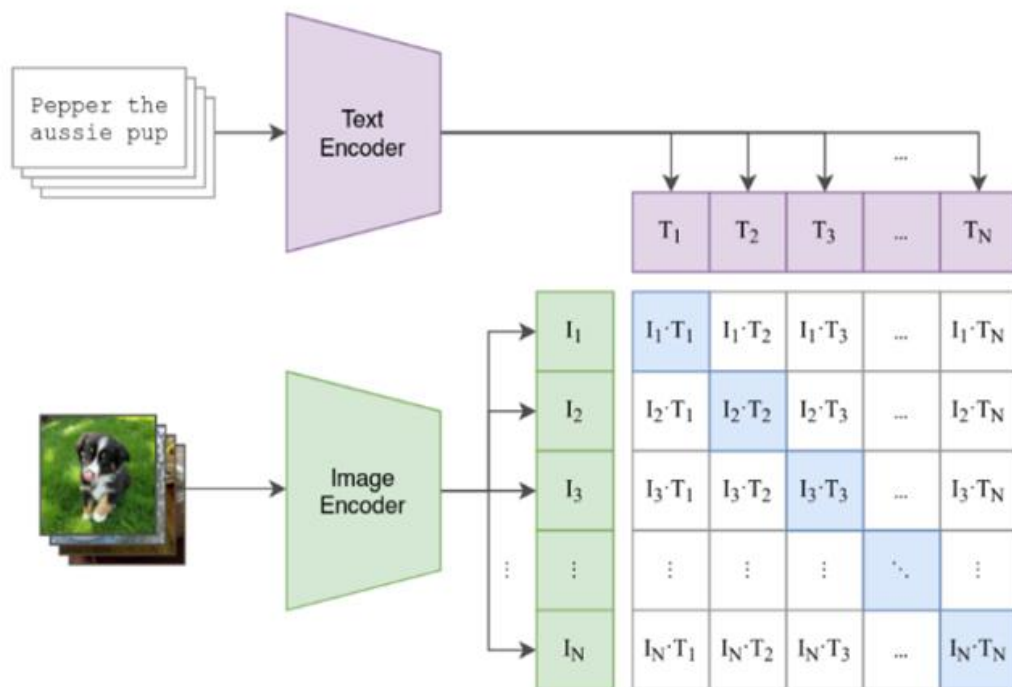
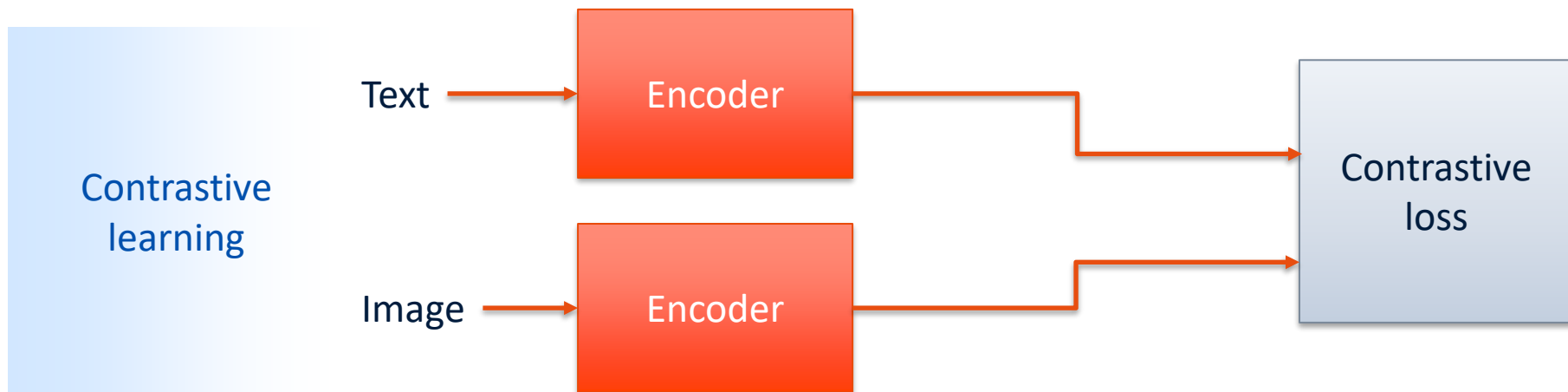
“a dog jumping through the snow”

$768 \times 49 = 37632$

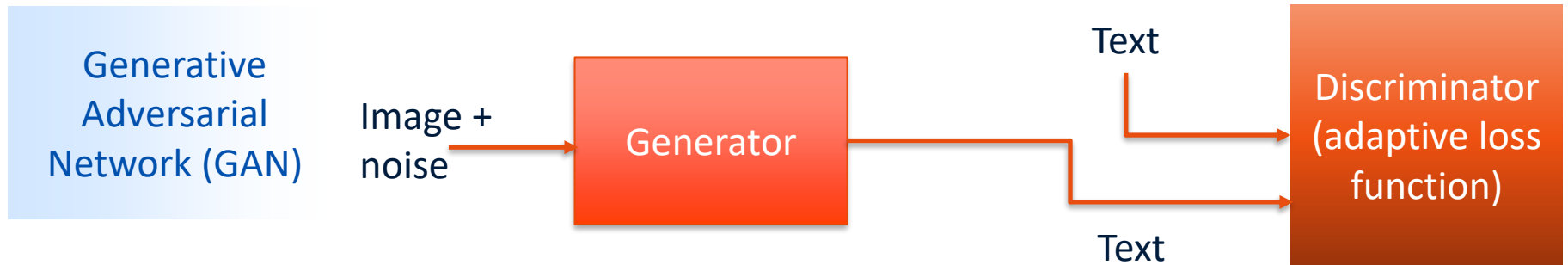
VQA: decoder-only formulations (with tokenizer)



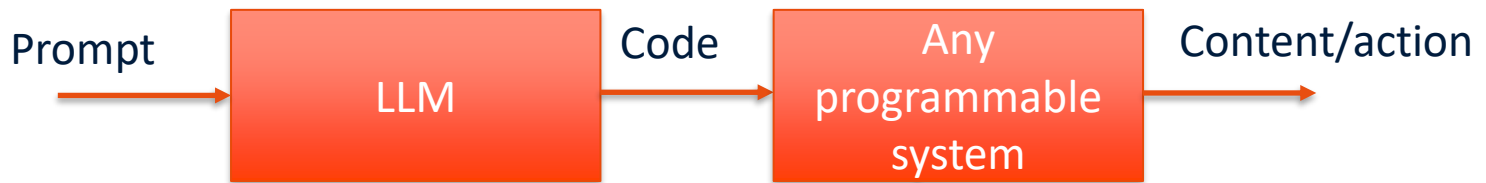
Contrastive formulations



GANs – Generative Adversarial Networks



Action/data through code generation



Programmable systems

- Other generative models
- Robots
- Vehicles
- Integrated circuit design systems
- Industrial machinery
- Social media
- Financial institution and markets
- Architecture, interior or product design

Example from our MSc projects*

- Generate CAD** code representing an exhibition stand (*for a specialized company that delivers thousands of exhibition stands / year*)
- Input:
 - Floorplan
 - Text description of requirements
- Output:
 - Stand description in CAD language
 - CAD -> visuals and budget



* Johannes Coppeneur, DACS MSc Thesis, Sept. 2025

** CAD – Computer Aided Design (a variant of the LISP language)

Contrastive Language Image Pre-training (CLIP)

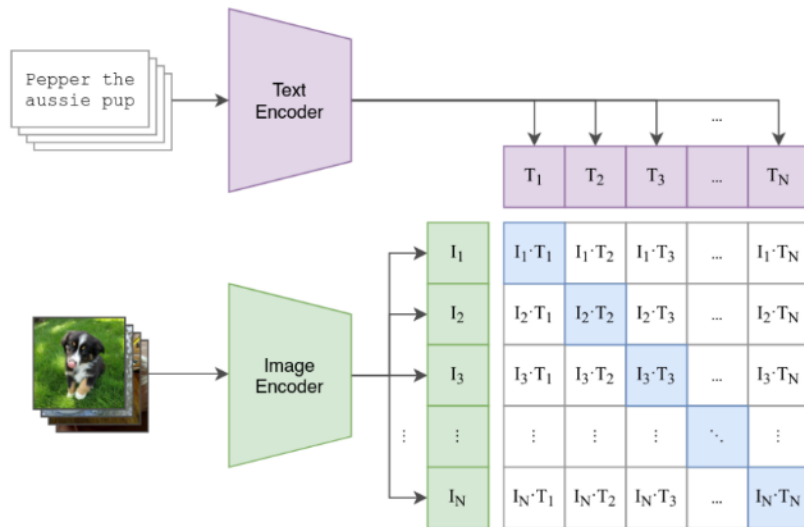


CLIP Image-to-text (Contrastive Learning)

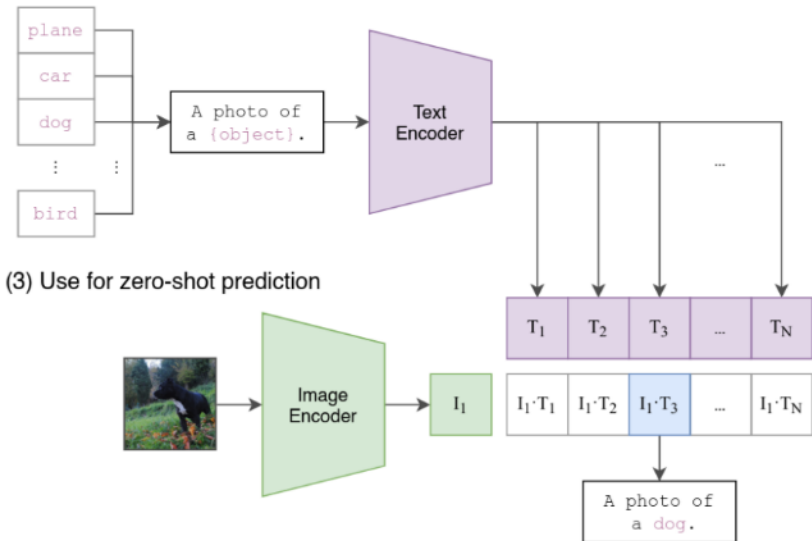
Learning Transferable Visual Models From Natural Language Supervision

2

(1) Contrastive pre-training



(2) Create dataset classifier from label text



(3) Use for zero-shot prediction

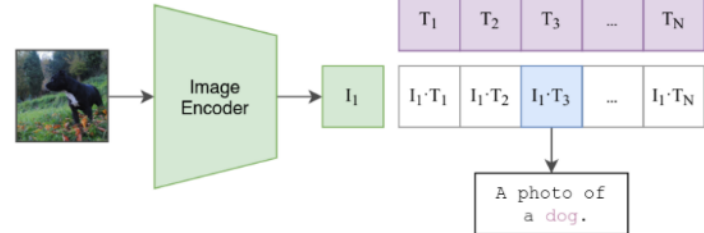


Figure 1. Summary of our approach. While standard image models jointly train an image feature extractor and a linear classifier to predict some label, CLIP jointly trains an image encoder and a text encoder to predict the correct pairings of a batch of (image, text) training examples. At test time the learned text encoder synthesizes a zero-shot linear classifier by embedding the names or descriptions of the target dataset's classes.

Trained using 400M
image/text pairs

[\[2103.00020\] Learning Transferable Visual Models From Natural Language Supervision \(arxiv.org\)](#)

Image-to-text (Contrastive Learning)

Positive pairs

a white dog in a jump holding a frisbee in his mouth



a woman and child laugh



a large brown dog is crouching in a grassy field



Negative pairs

a dog running on the sand by the beach



three people with backpacks walk up a road



a baby boy vacuuming his house and carpet



Contrastive Learning: image & text example

- InfoNCE loss function
$$L_i = -\log \frac{\exp(z_i^T w_i / \tau)}{\sum_{j \neq i}^J \exp(z_i^T w_j / \tau)}$$

“a white dog
in a jump
holding a
frisbee in his
mouth”

Text encoder

z_i



Image
encoder

w_i

a dog running on
the sand by the
beach

z_k



w_k

three people with
backpacks walk
up a road

z_l



w_l

a baby boy
vacuuming his
house and carpet

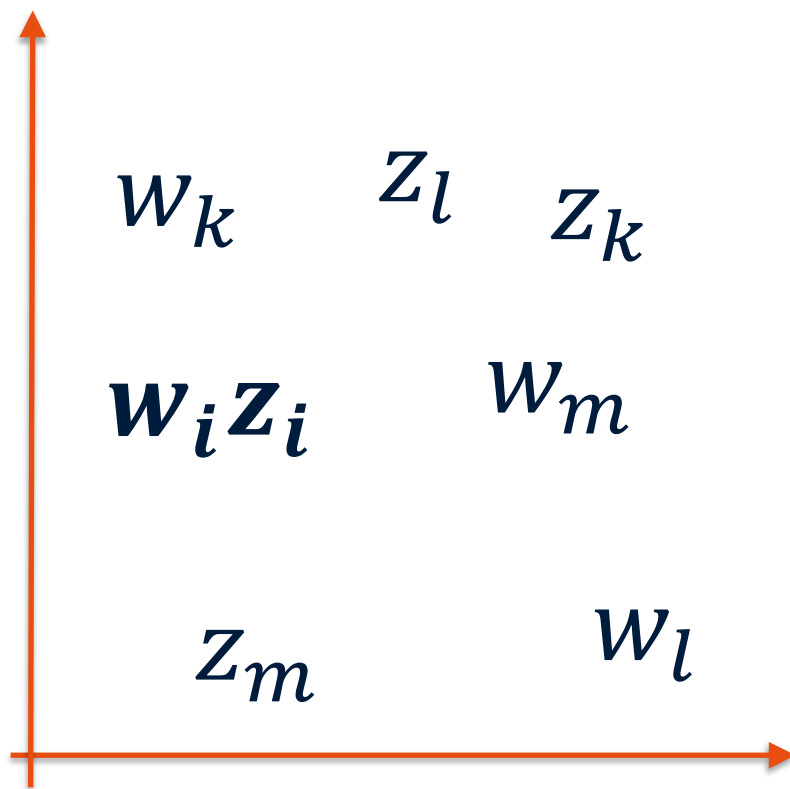
z_m



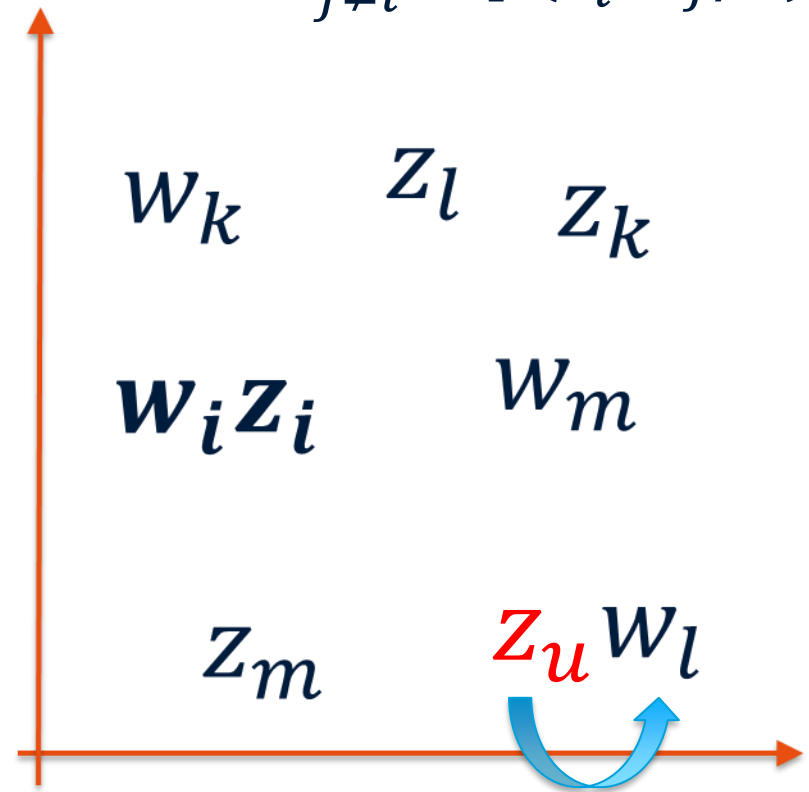
w_m

Contrastive Learning: image & text example

- InfoNCE loss function
$$L_i = -\log \frac{\exp(z_i^T w_i / \tau)}{\sum_{j \neq i}^J \exp(z_i^T w_j / \tau)}$$

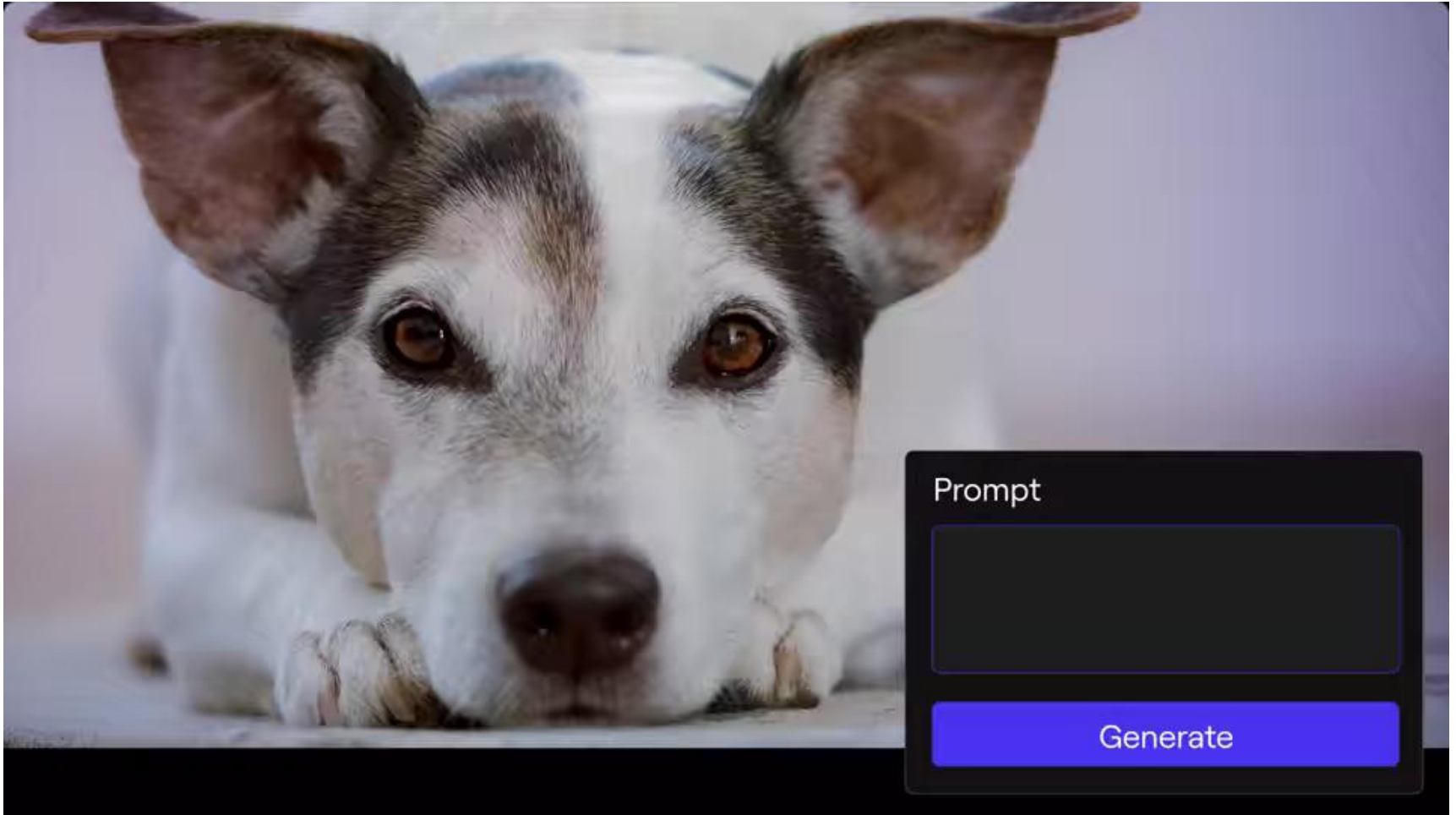


After training

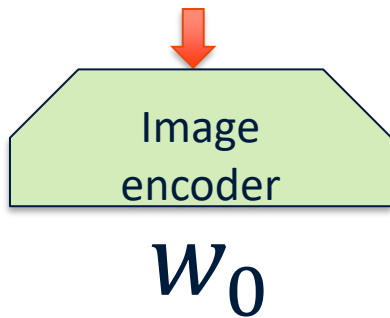


During inference

Mix vectors Z_u , W_i and use diffusion methods



CLIP example*



"two cats on a couch"

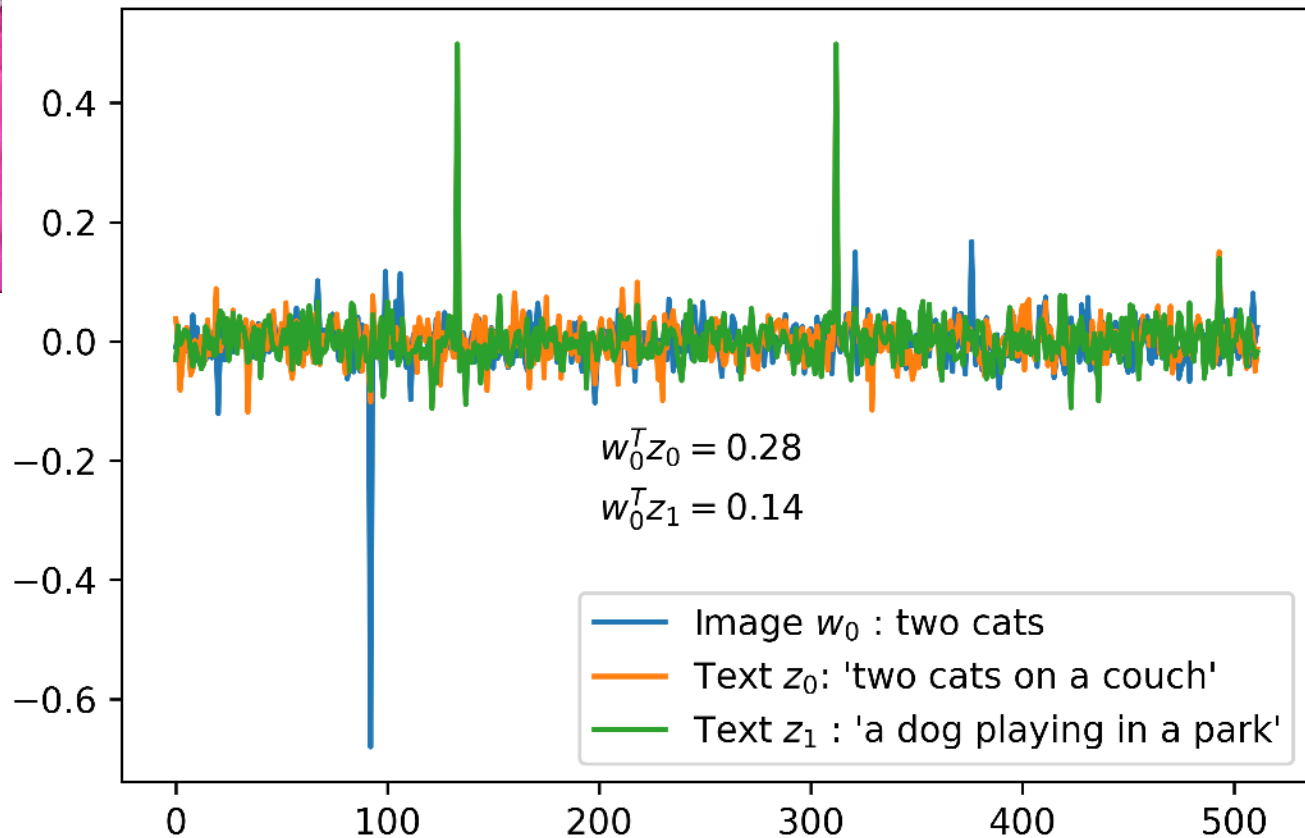
"a dog playing in a park"



Text encoder

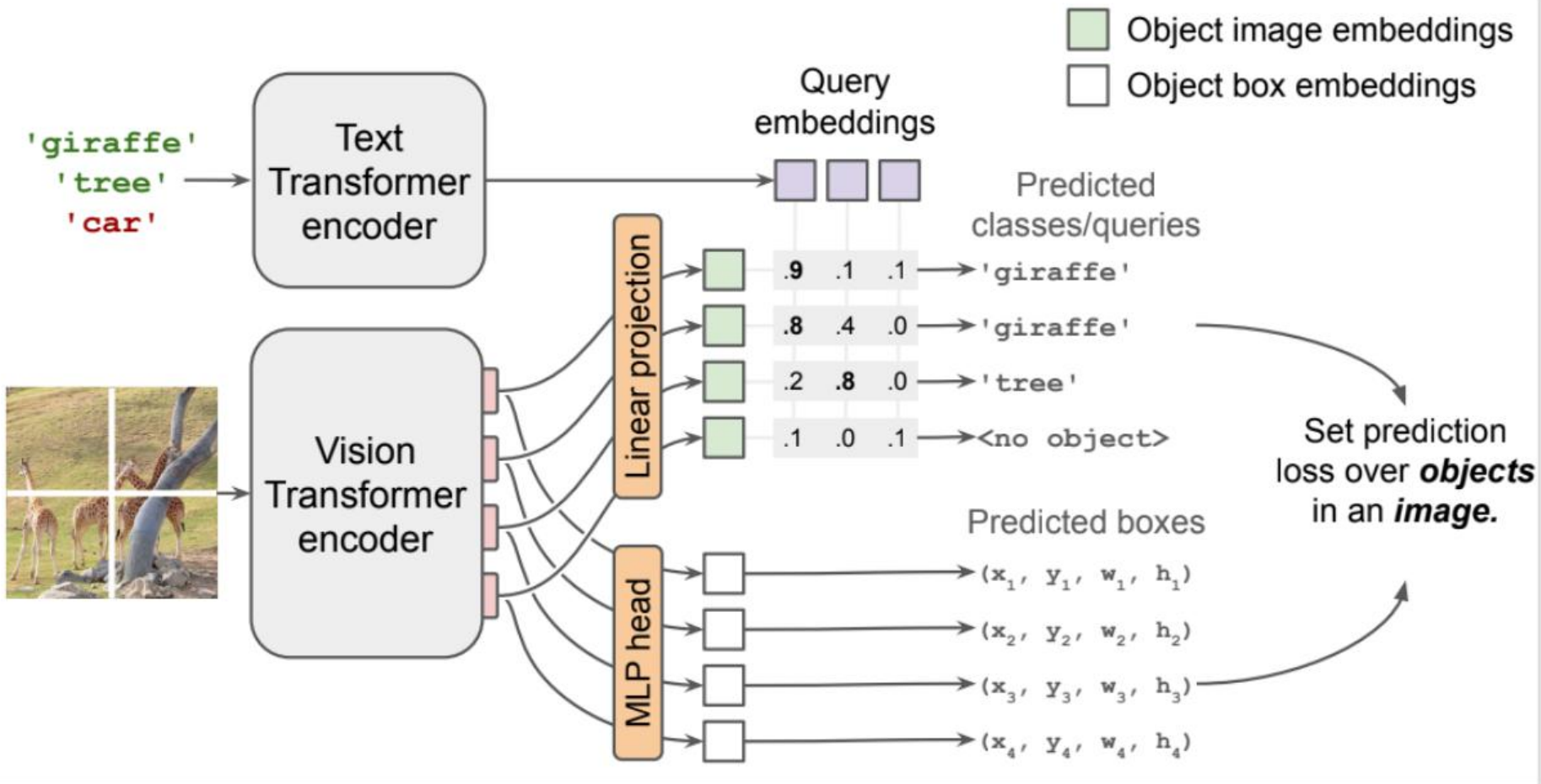
z_0

z_1



*openai/clip-vit-base-patch32 in Huggingface

Open-vocabulary image detection



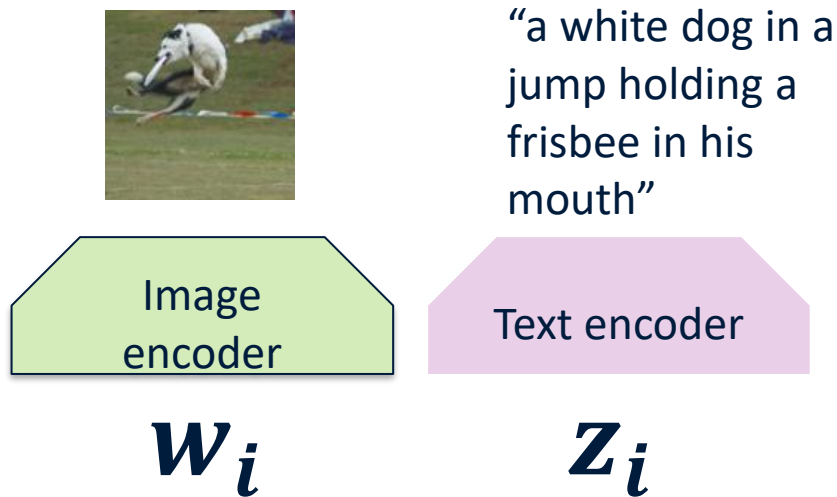
Open-vocabulary image detection

Enter comma-separated queries:

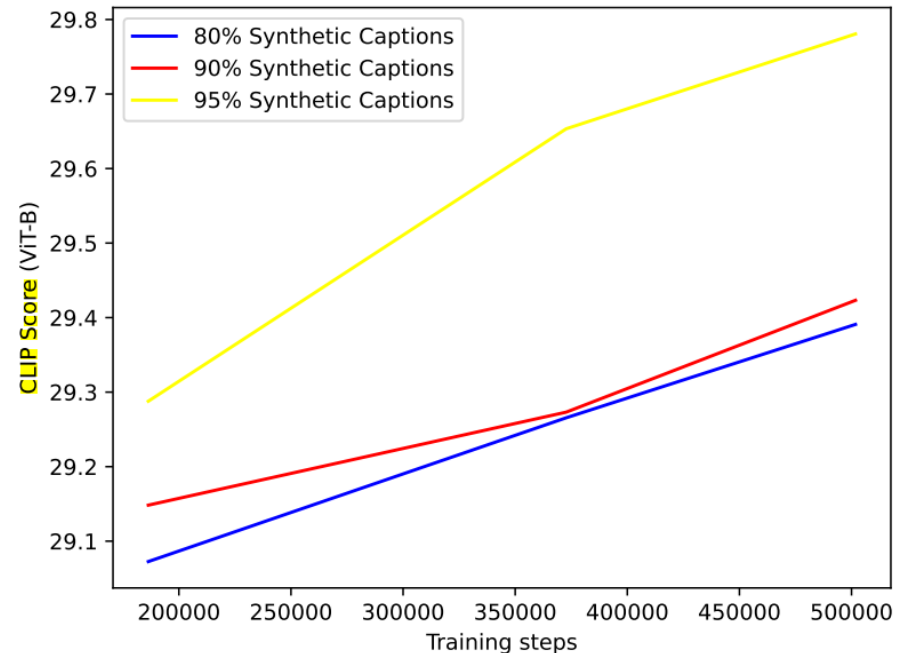


CLIP Score – measure for image-text models

- Becoming a popular model for text-to-image, image-to-text tasks



$$\text{The CLIP score} = \frac{w_i^T z_i}{|w_i| |z_i|}$$



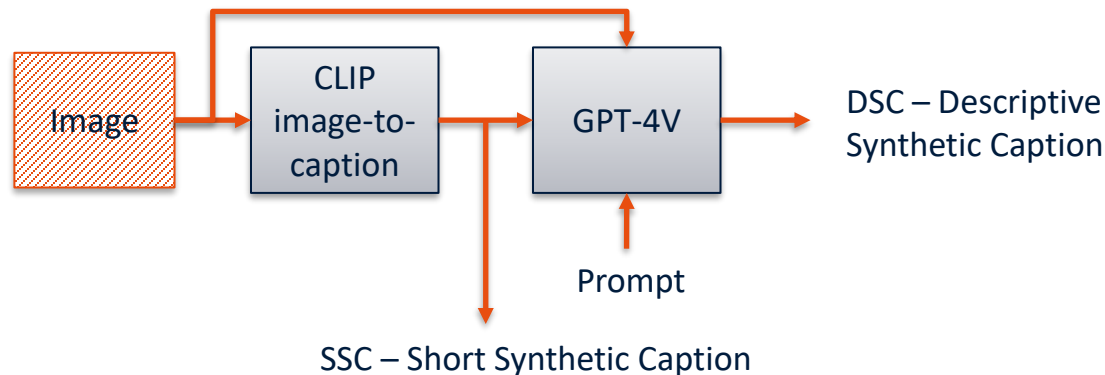
DALL-E 3 performance report

Synthetic data and data augmentation, optimization in multimodal models



Fusion of synthetic and real data in training

- Open Internet image content has often bad captions (alt-text field)
- “Our data engine, instead of relying on labor-intensive manual annotation, autonomously generates a comprehensive visual dataset called FLD-5B, encompassing a total of 5.4B annotations for 126M images...” (Florence-2, Gemini?)
- DALL-E 3 uses generated image captions in training data to replace original captions: “..a mixture of 95% synthetic captions and 5% ground truth captions”*



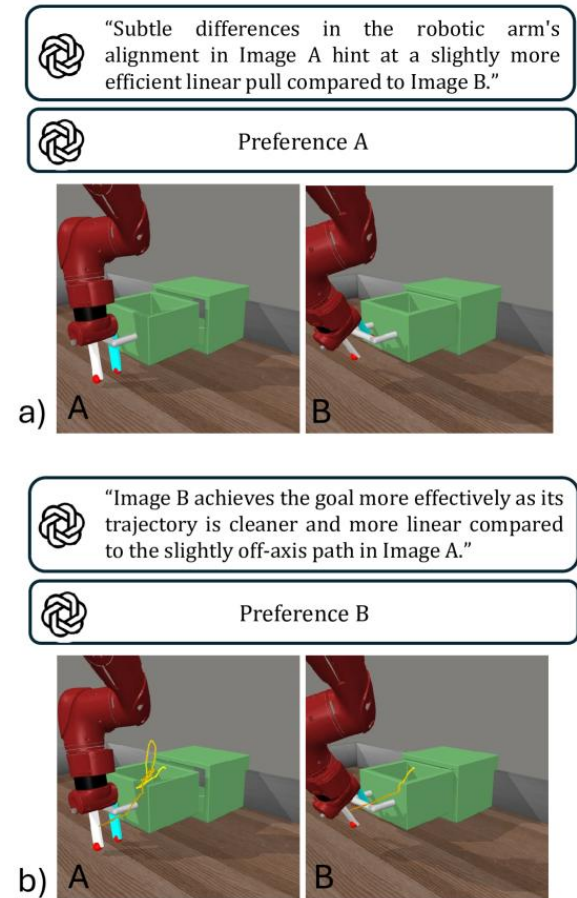
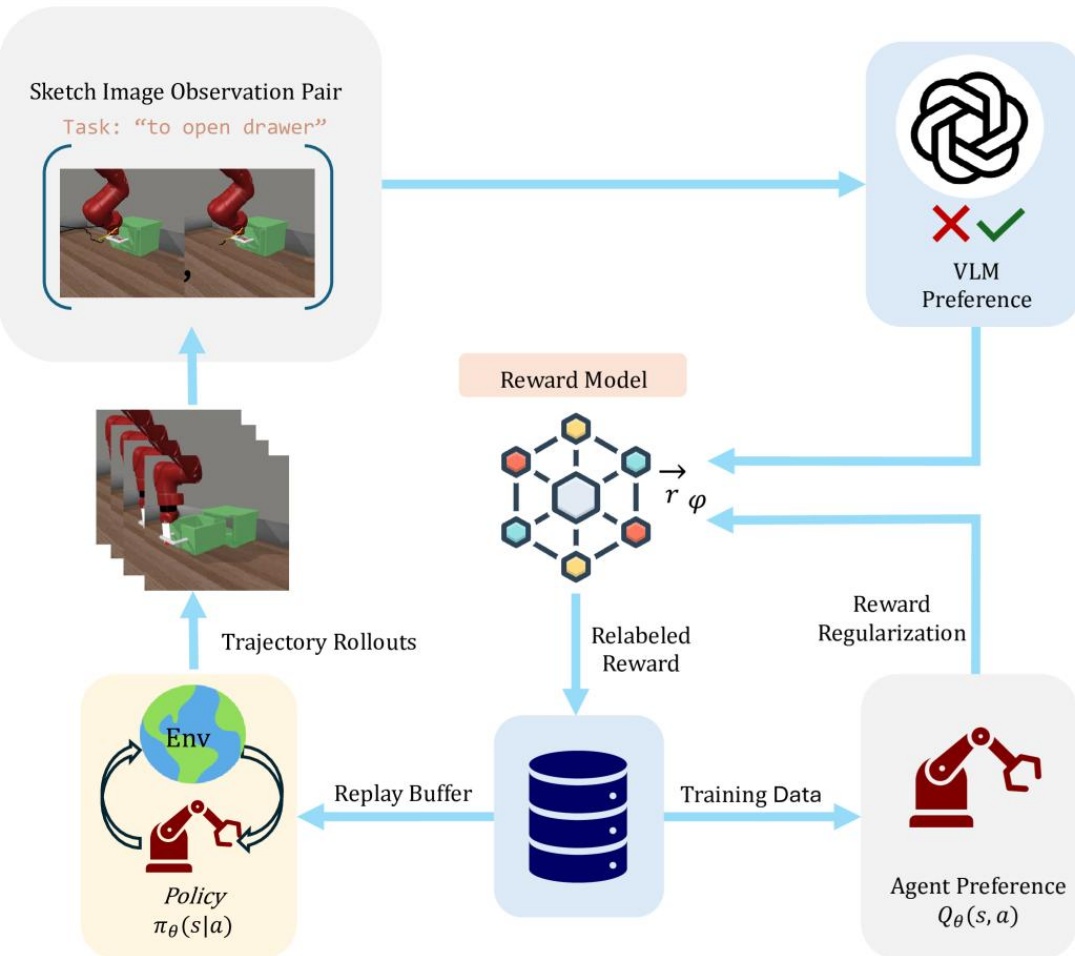
[*dall-e-3.pdf \(openai.com\)](#)

DALL-E 3 data *recaptioning*

Image			
Image			
	Alt Text	now at victorian plumbing.co.uk	is he finished...just about!
	SSC	a white modern bathtub sits on a wooden floor.	a quilt with an iron on it.
DSC			23 (19 of 30) 1200
		this luxurious bathroom features a modern freestanding bathtub in a crisp white finish. the tub sits against a wooden accent wall with glass-like panels, creating a serene and relaxing ambiance. three pendant light fixtures hang above the tub, adding a touch of sophistication. a large window with a wooden panel provides natural light, while a potted plant adds a touch of greenery. the freestanding bathtub stands out as a statement piece in this contemporary bathroom.	a jar of rhubarb liqueur sitting on a pebble background. rhubarb pieces in a glass jar, waiting to be pickled. the colors of the rhubarb range from bright red to pale green, creating a beautiful contrast. the jar is sitting on a gravel background, giving a rustic feel to the image.

Figure 3 – Examples of alt-text accompanying selected images scraped from the internet, short synthetic captions (SSC), and descriptive synthetic captions (DSC).

VLA model optimized by RL



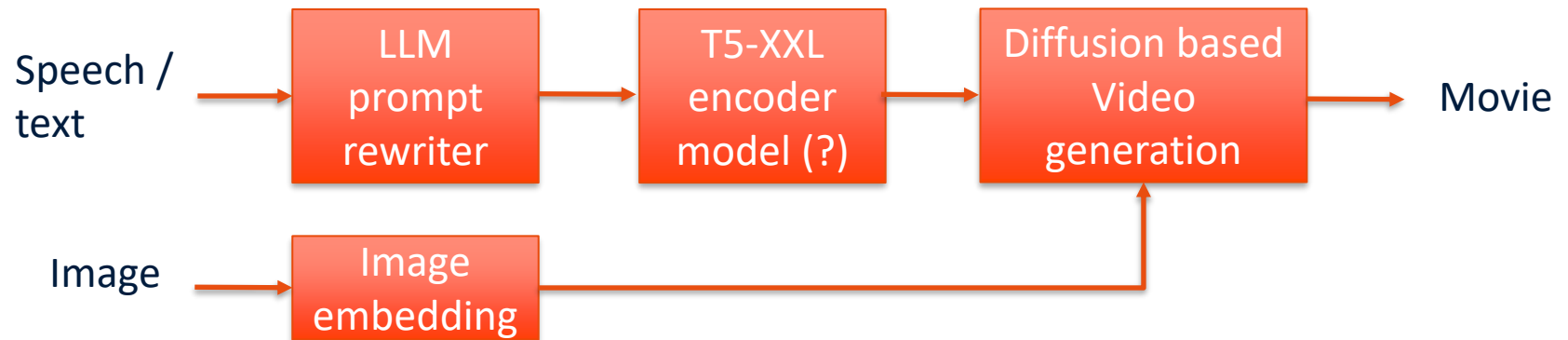
[VARP: Reinforcement Learning from Vision-Language Model Feedback with Agent Regularized Preferences](#)



A movie of a group of ducks dancing around of a large statue forming the letters of the text "I <heart> DACS" where <heart> is a symbol for heart. Background shows the landscape around Maastricht and the Maas river and people are cheering. Example with Google Veo 3.1

Example: Google DeepMind Veo 3.1

(architecture based on available information)



- LLM-based prompt rewriter provides an additional “thinking” step

Capabilities of Multimodal Models



“Chain-of-Frames” – multimodal thinking

- The same as chain-of-thought but for video

Prompt: The background changes to white.

Then:

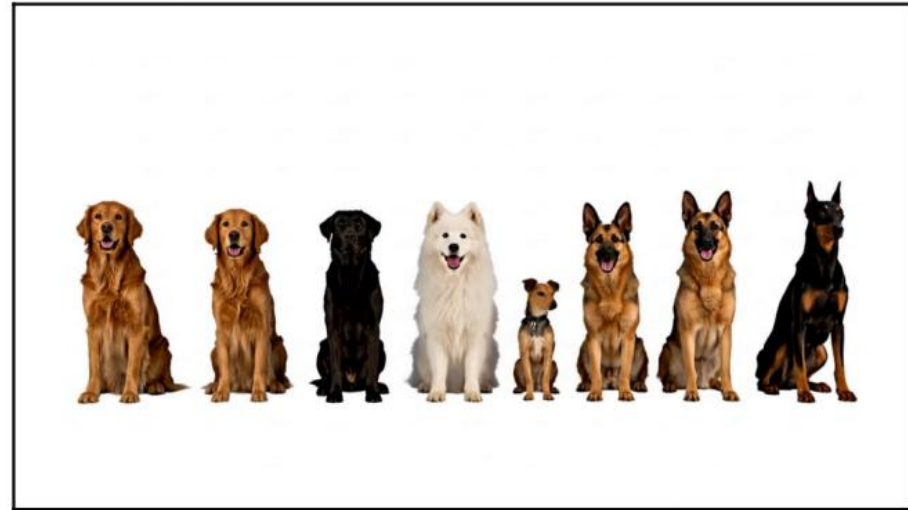
If there is just a single animal: the animal sits in the middle of the image, looking straight at the camera.

If there are multiple animals: all animals line up in a row, with ample white space between them.

Original image



Generated frame (Veo 3)



Video Understanding



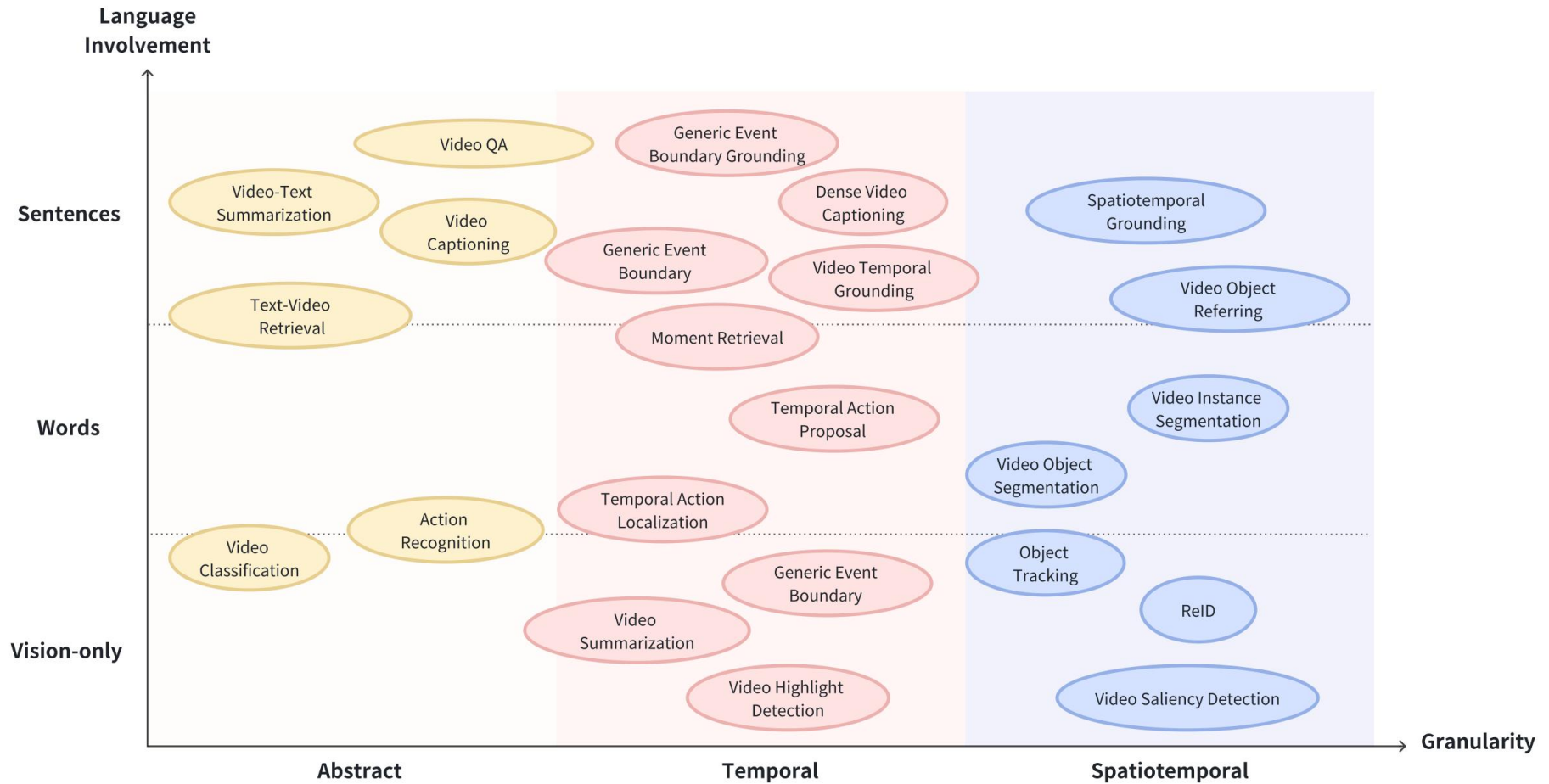
Dense Video Captioning

A cyclist performs a wheelie across a narrow log obstacle in a wooded outdoor setting, demonstrating balance and precision on a mountain bike. The scene is framed by bare trees and fallen leaves, suggesting an autumn day, with the rider's focus on maintaining control across the uneven terrain.

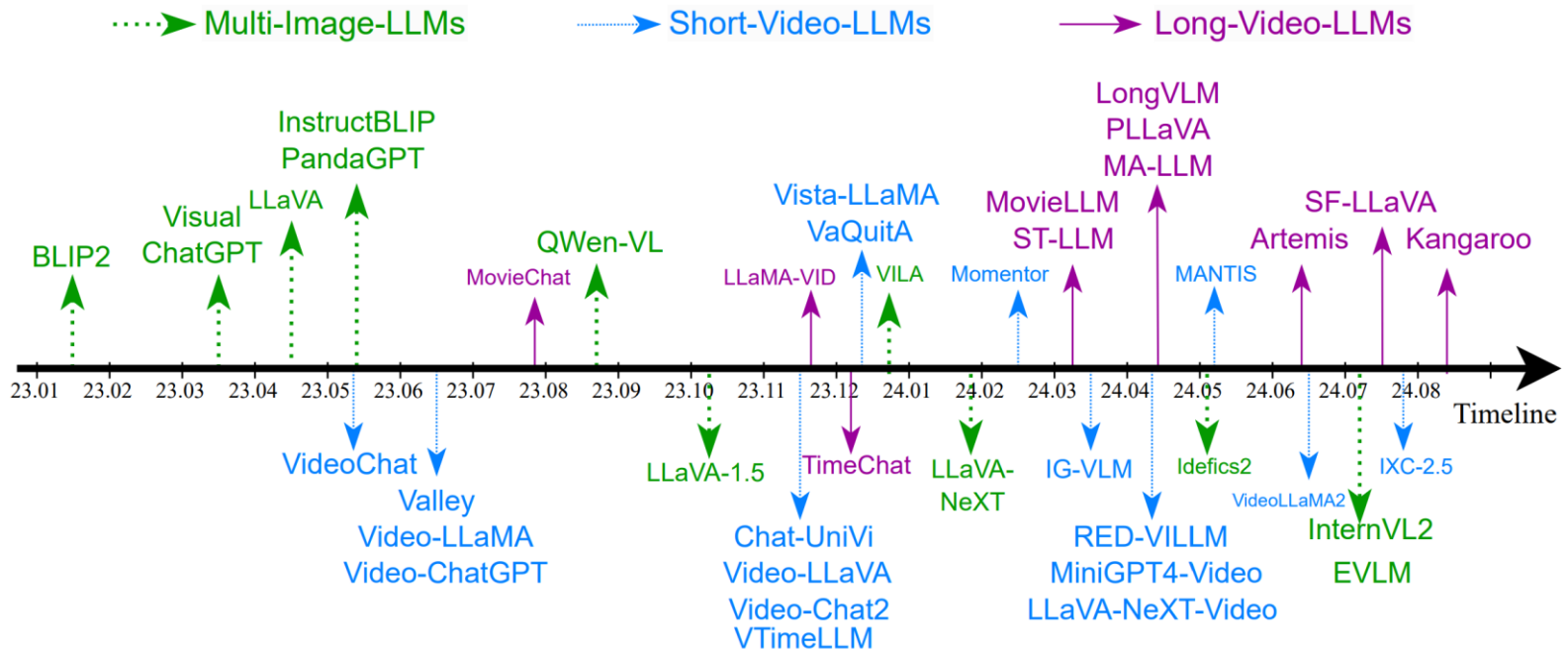
Single Sentence Captioning

A cyclist skilfully rides across a narrow log in a forested area

Video Understanding



Video Understanding



Transformer – a versatile model for various data types



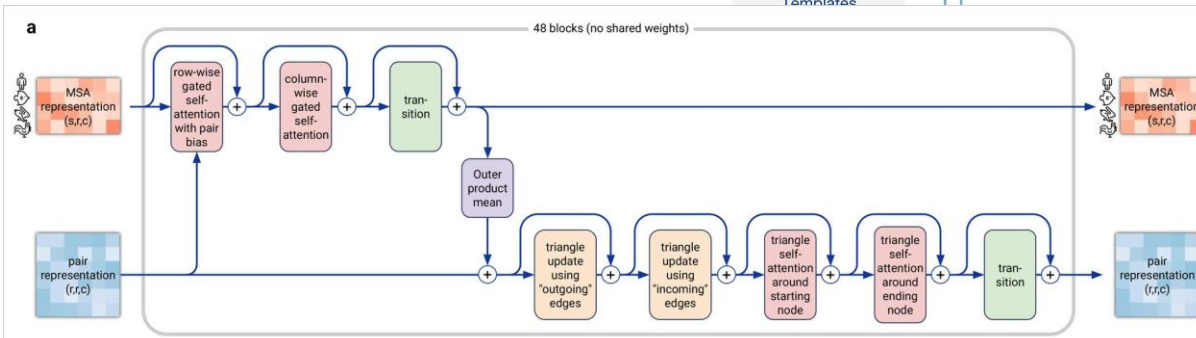
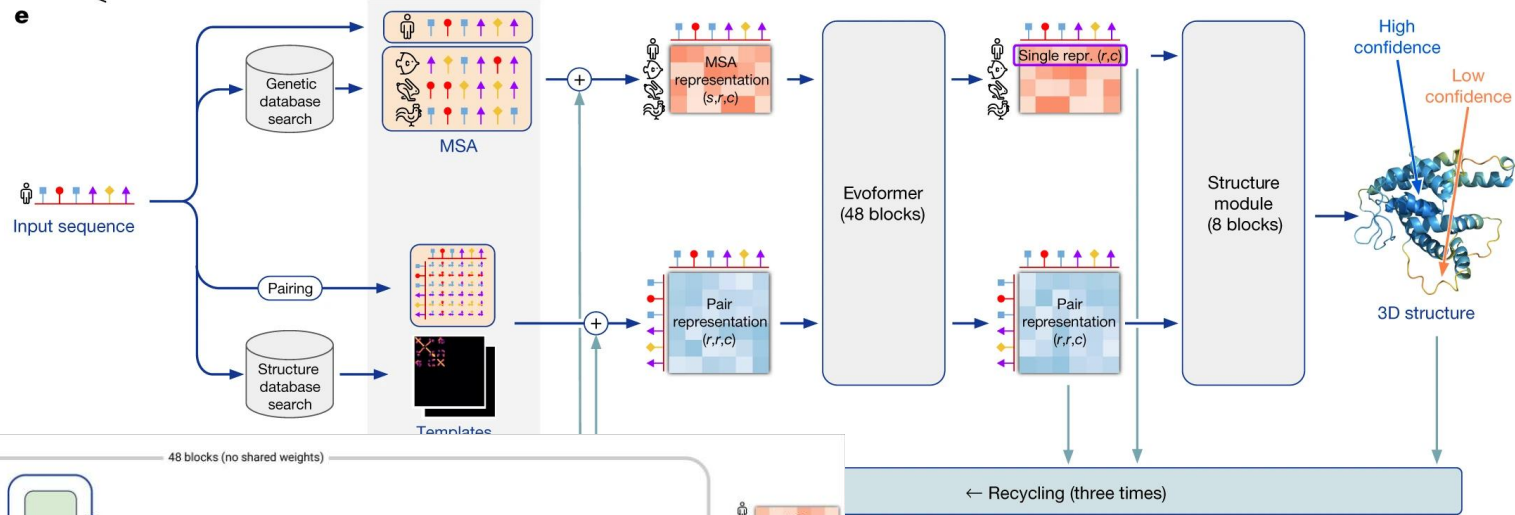
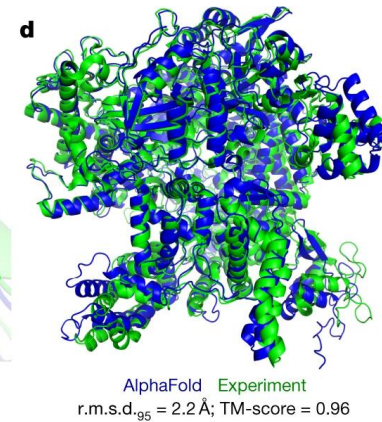
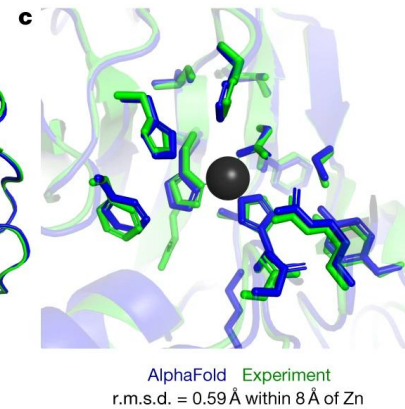
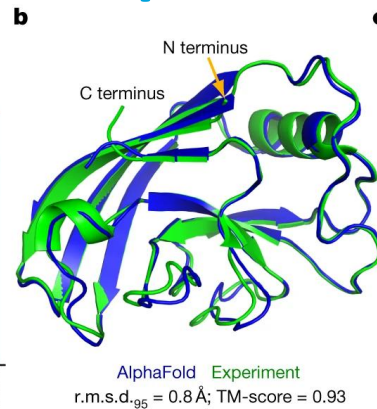
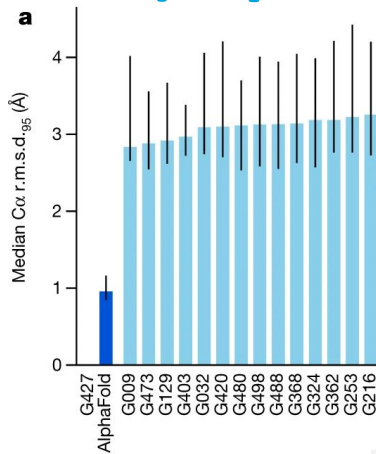
More



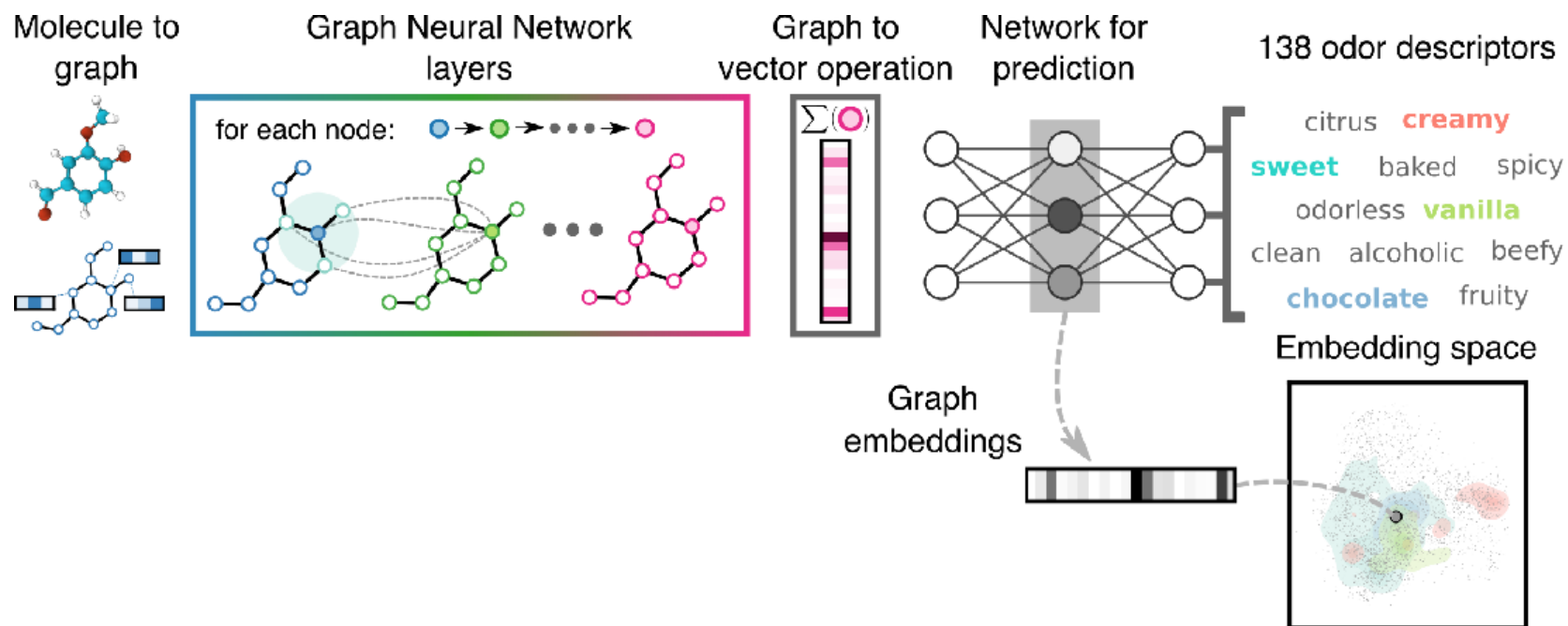
Protein data (AlphaFold)

Won Nobel Prize for Chemistry in 2024!

Protein folding:
linear chain of amino acids to a complex 3D structure



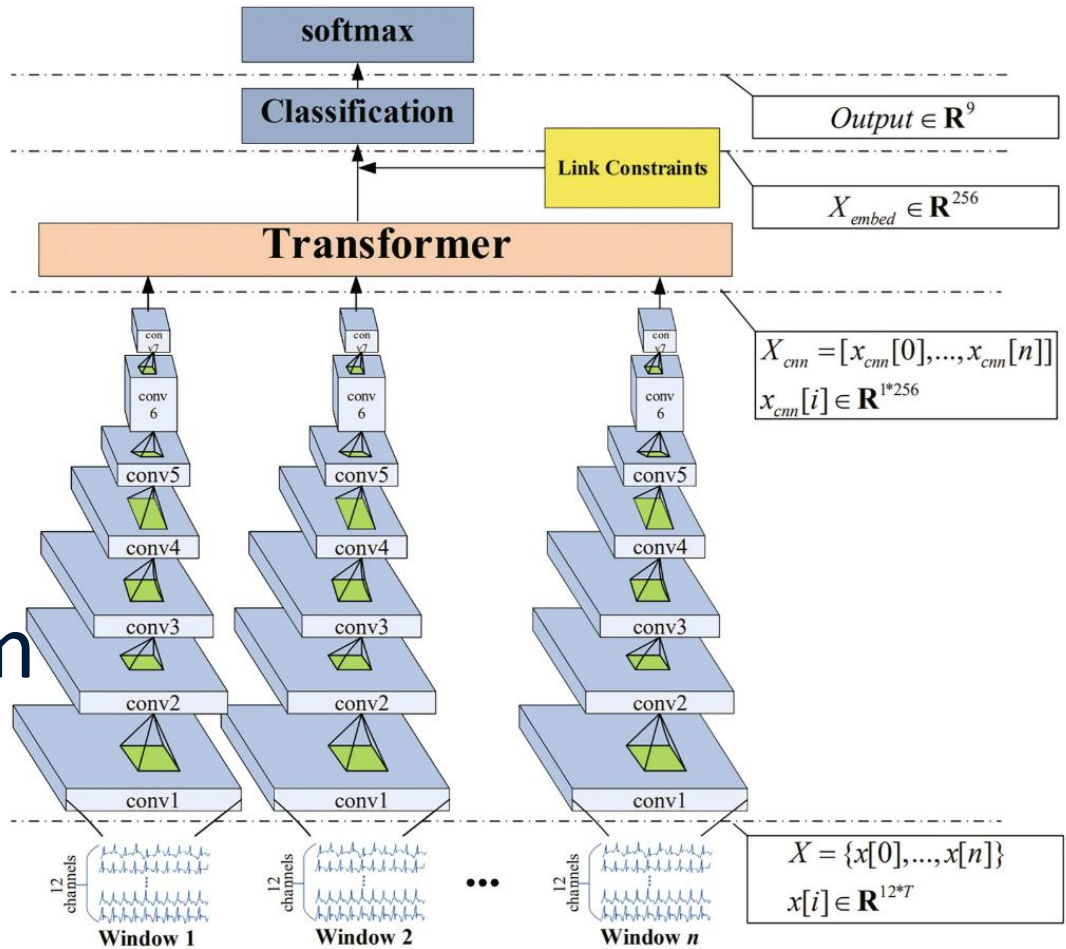
Scent (olfaction)



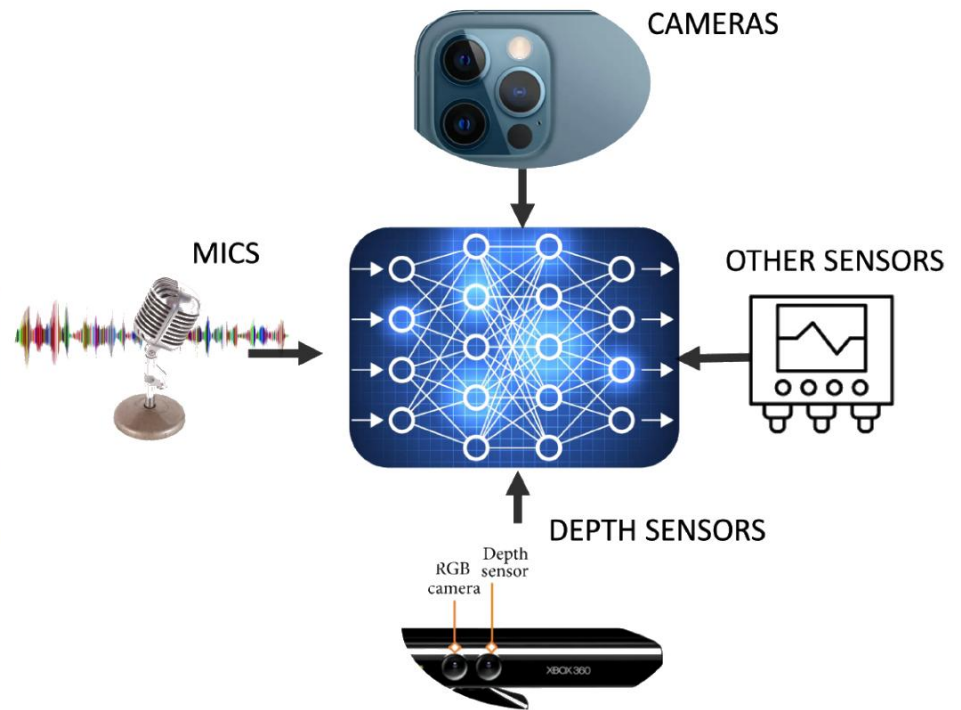
Cardiology – Electrocardiogram (ECG)

A model for the classification arrhythmias from 12-lead electrocardiogram recordings

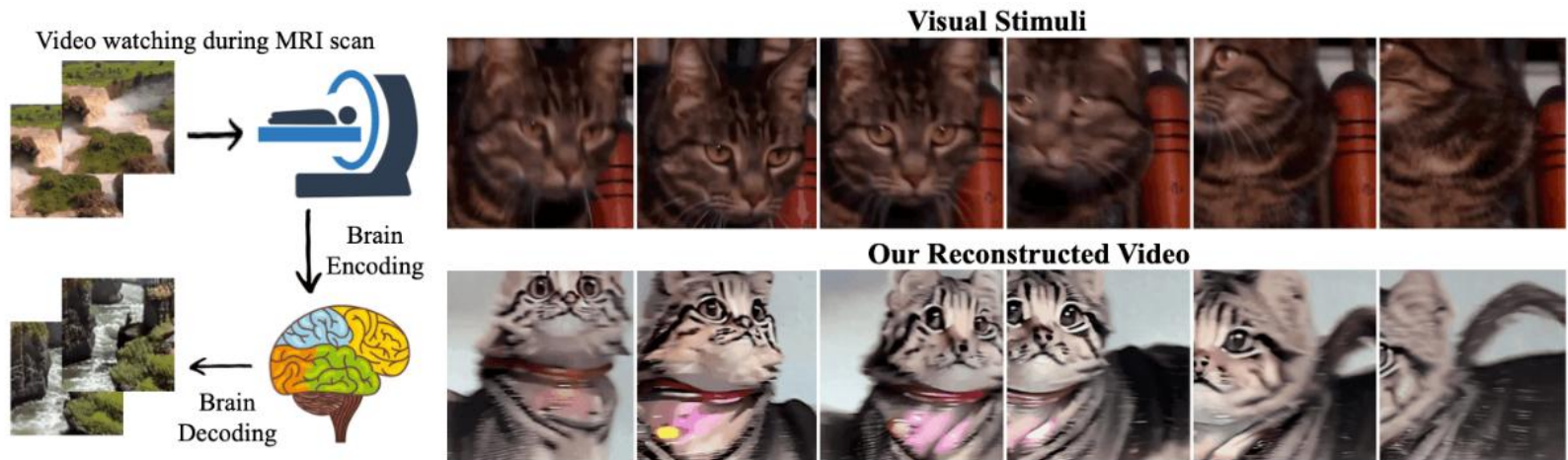
From: [Constrained transformer network for ECG signal processing and arrhythmia classification](#)



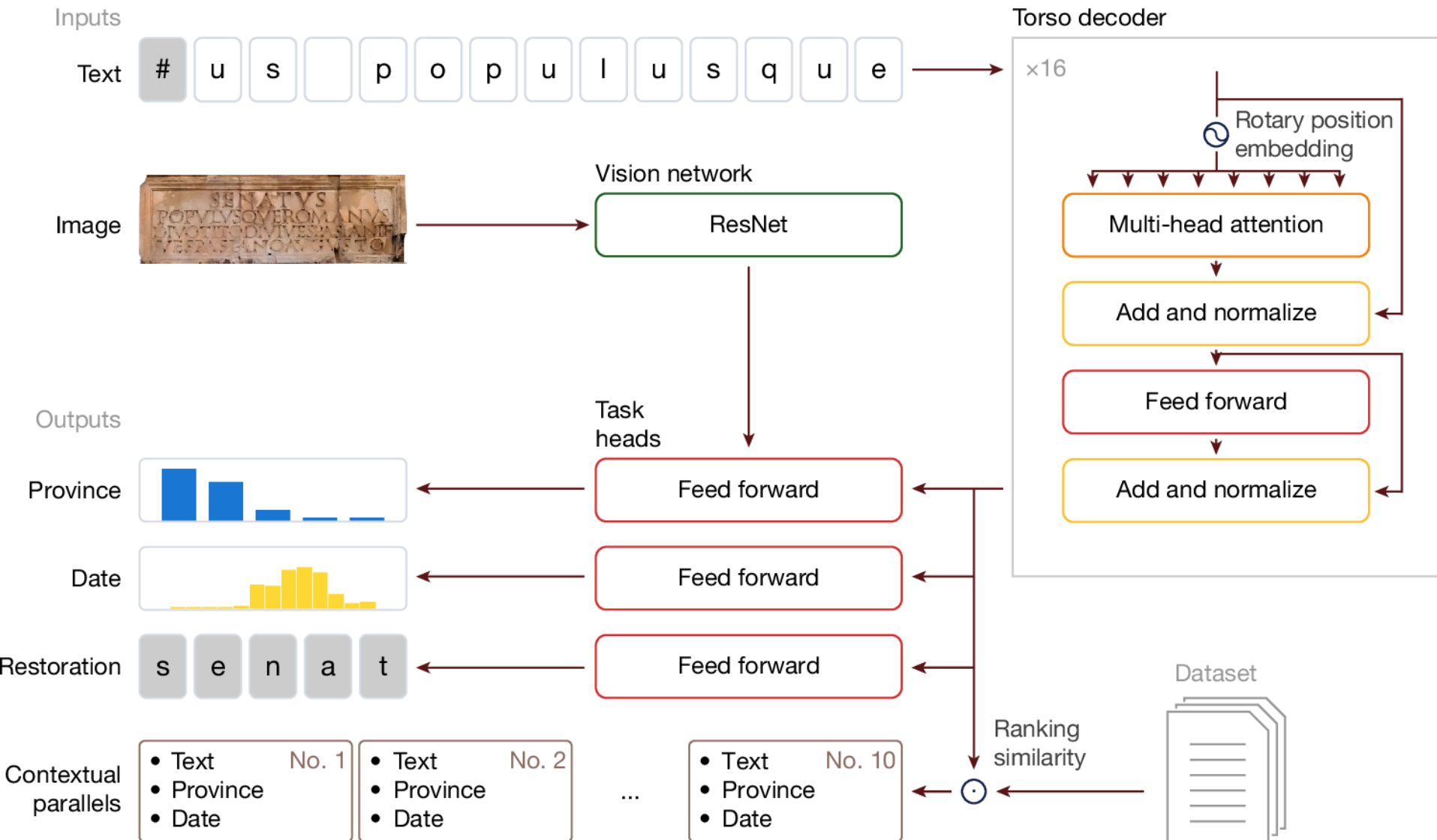
Robots are also multimodal



fMRI -> Video



Contextualizing ancient texts



“Multimodal prompting”

Input Image	Input Audio (transcribed)	Model Response: Text
	 What's the first step to make a veggie omelet with these ingredients?	Crack the eggs into a bowl and whisk them.
	 Thank you for the instructions. I started making my omelet, does it look ready now?	It looks like it's almost ready. You can flip it over to cook the other side.
(No image - it's a follow up on the previous question)	 Why is it not ready?	It's not ready because the eggs are still runny.

Unified-IO

Image Editing



Remove the dock



Paint this image like Van Gogh



Render a sunset

Reference Image Generation




Generate an image of a car with the model in the first image and the color in the second image.



Image Generation

Generate an image of an astronaut riding a horse in the forest. There is a river in front of them with water lilies.



Generate an image of an elephant swimming underwater. aesthetic. Fantasy.



Multiview Image Completion




Add the missing details to the masked image (left) using the reference image (right).



Free Form VQA



What food could you make with these ingredients? Give me the recipe.

One delicious recipe using these ingredients is chocolate pudding! Here's the recipe:

Ingredients:
- 1 cup all-purpose flour, - 1/2 cup sugar

Instructions:
1. In a large bowl whisk together the flour, sugar..
2. In a separate bowl, mix together the eggs ...

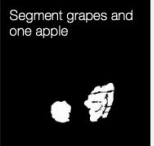
Visual Parsing & Segmentation



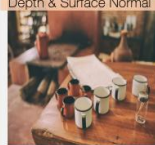
Which fruits are in this image? List them in json format with the name of the fruit as the key and the color of the fruit as the value.

```
{
  "banana": "Yellow",
  "apple": "Red",
  "grapes": "Green"
}
```

Segment grapes and one apple



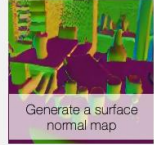
Depth & Surface Normal



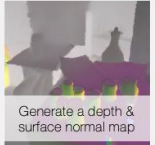
Generate a depth image



Generate a surface normal map



Generate a depth & surface normal map



Keypoint Estimation



Find the visible keypoints corresponding to the person located in the highlighted region.



Visual based Audio Generation



Generate an audio track for this band.




Generate an audio track for this band.



Visual Audio Localization



(drum sounds)

Identify the locations of the instruments producing the given sound.



Robotic Manipulation

= blicket
= dax
is kobar than
Put the less kobar blicket into the dax.





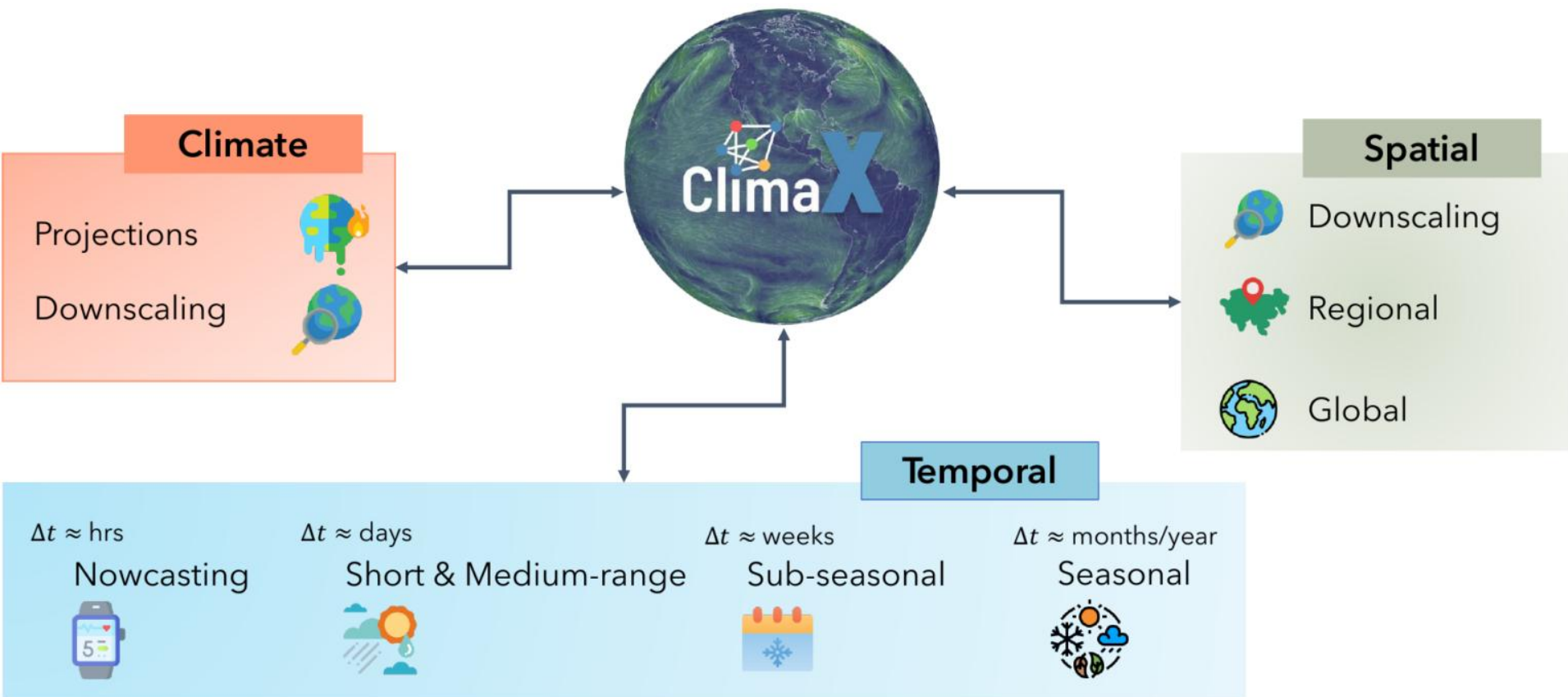
Future Frame Prediction

Given the initial image and a sequence of actions, predict the next frames

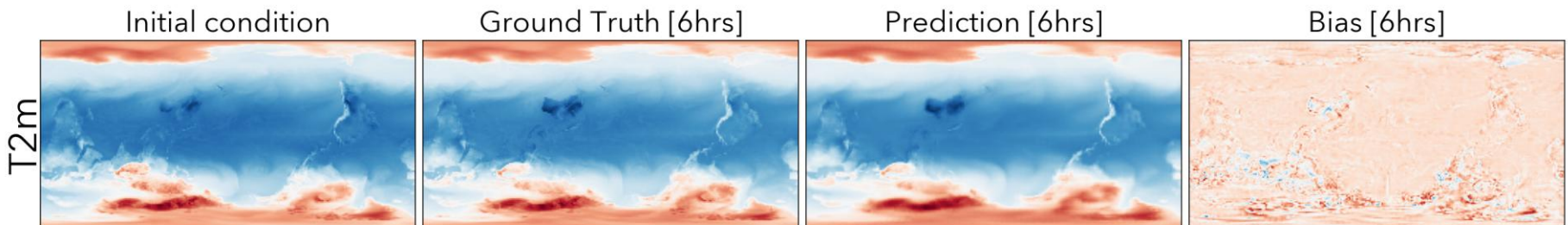
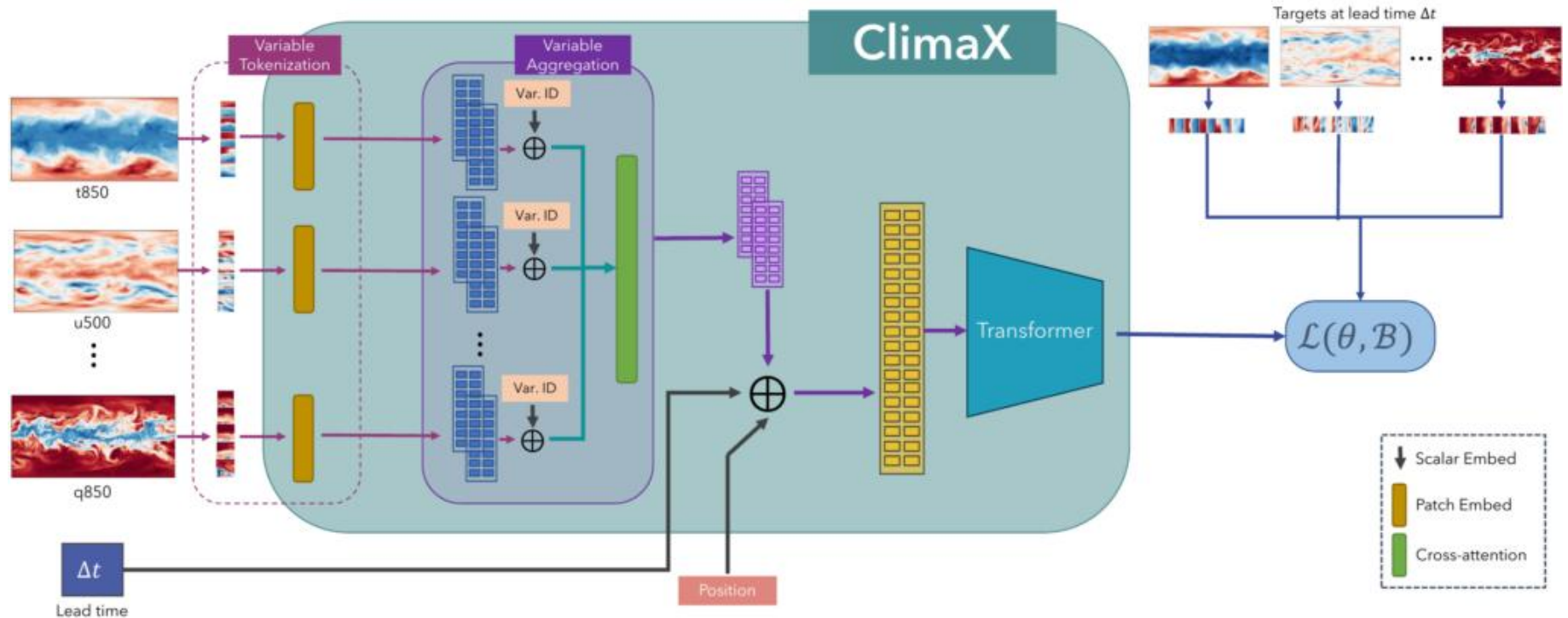
Generated Images



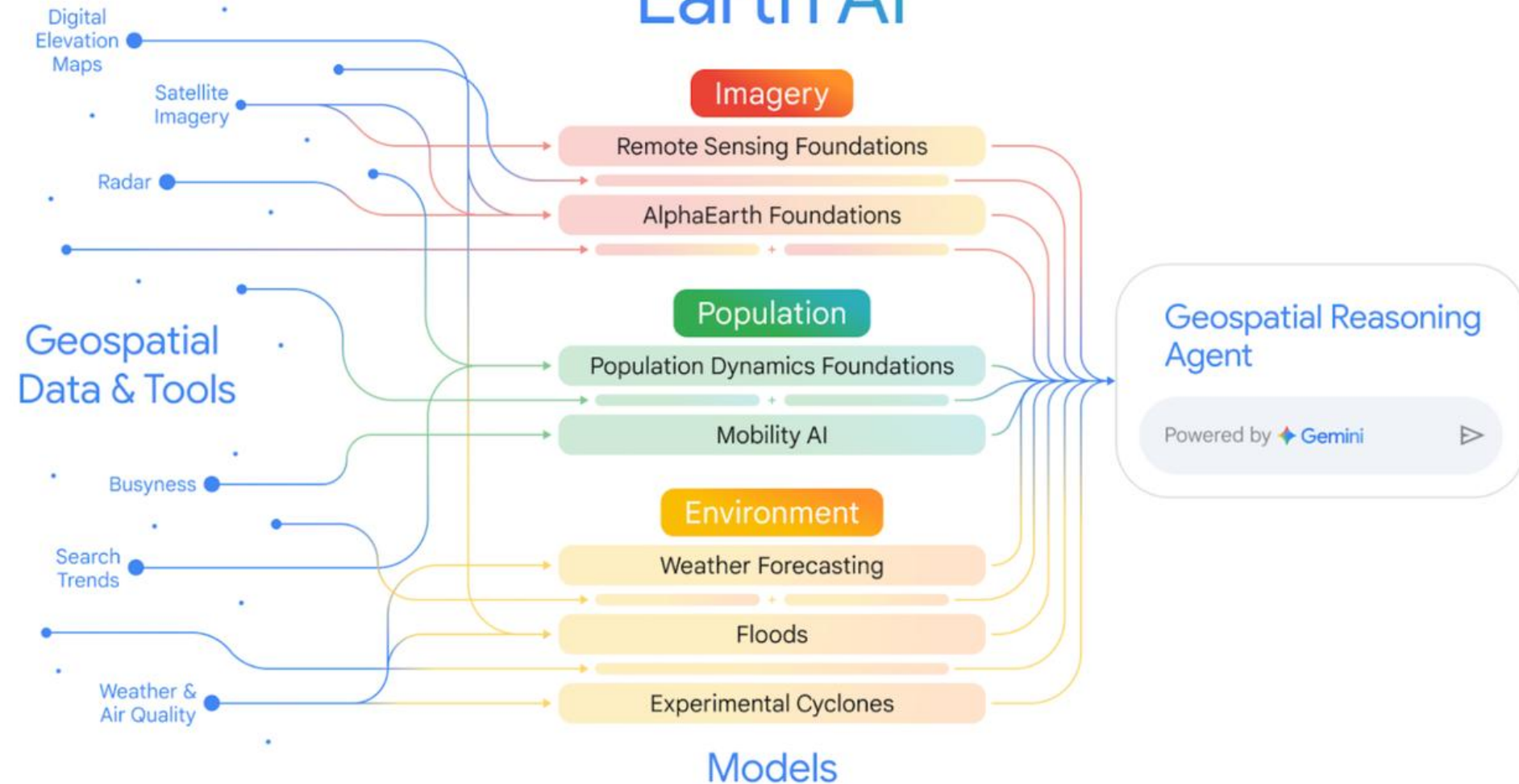
Distributional Graphformer for weather



Distributional Graphformer for weather



Earth AI



- [Google Earth AI: Unlocking geospatial insights with foundation models and cross-modal reasoning](#)

Questions?



Maastricht University

Exercise 1: Contrastive Vision-Language Learning and InfoNCE Loss (CLIP)

CLIP (Contrastive Language-Image Pre-training, Radford et al., 2021) jointly trains an image encoder and a text encoder to project both modalities into a shared embedding space using a contrastive objective.

- The lecture presents the symmetric InfoNCE contrastive loss for a matched image-text pair (z_i, w_i) :

$$\mathcal{L}_i = -\log \frac{\exp(z_i^T w_i / \tau)}{\sum_{j=1}^J \exp(z_i^T w_j / \tau)}$$

Define each term in this equation (including the temperature parameter τ , the positive pair dot product $z_i^T w_i$, and negative pairs $z_i^T w_j$). What geometric effect does minimizing this loss have on the orientation of image and text vectors in the shared latent space?

- The lectures also illustrates zero-shot classification with CLIP using prompt engineering (e.g., matching image embedding w_0 against prompt texts $z_0 = \text{"two cats on a couch"}$ and $z_1 = \text{"a dog playing in a park"}$). Explain step-by-step how CLIP performs zero-shot prediction over arbitrary image classes without training a task-specific classification head.
- The lecture demonstrates that the CLIP score can also serve as an automated evaluation metric. How is the cosine similarity between generated images and text captions used to evaluate generative models like DALL-E 3?

Exercise 2: Multimodal Architecture Formulations (Encoder-Decoder vs. Autoregressive Decoder-Only)

- The lecture slides distinguish between two primary architectural paradigms for Vision-Language Models (VLMs): encoder-decoder cross-attention networks and unified decoder-only autoregressive models.
 - In the **Encoder-Decoder** image captioning formulation, how is the visual input processed into patch/token representations, and how does the text decoder ingest these representations via cross-attention?
 - In the **Decoder-Only** formulation for Visual Question Answering, explain how image patch embeddings and tokenized text question tokens are fused before entering the autoregressive decoder. Why is the entire multimodal input treated as a single concatenated sequence of discrete/continuous tokens?
 - Contrast these pure neural foundation models with the classical **Symbolic Semantic Web / MPEG-7** approaches from early slides in the lecture. What are the trade-offs between symbolic, rule-based knowledge graphs (ontologies, RDF triples) and continuous neuro-symbolic grounded embeddings regarding explainability, maintenance effort, and handling of ambiguous visual scenes?

Exercise 3: Synthetic Data Recaptioning and Vision-Language-Action (VLA) Grounding

Recent advances in multimodal pre-training and downstream robotics rely heavily on synthetic data engines and reinforcement learning feedback.

- The lecture highlights that raw web-scraped image-text pairs often suffer from low-quality, noisy alt-text. Explain how **DALL-E 3** and **Florence-2** use automated recaptioning engines (e.g., Short Synthetic Captions vs. Descriptive Synthetic Captions generated by models like GPT-4V) to improve text-to-image generation fidelity. Why does training on up to 95% synthetic captions improve prompt adherence rather than causing model collapse?
- It also illustrates **VARP** (Vision-Language-Action models optimized by RL). How can a Vision-Language Model (VLM) act as an automated reward evaluator for robotic trajectory rollouts (e.g., scoring robot arm trajectories for the task "to open drawer")?
- We introduce the concept of "**Chain-of-Frames**" in generative video models like Google DeepMind Veo. How does generating intermediate visual frames parallel the step-by-step reasoning seen in textual Chain-of-Thought (CoT) prompting?

Discussion



Maastricht University

On Symbolic vs. Neural Grounding: Did the Semantic Web Lose to a Dot Product?

In the early 2000s, visionaries like Tim Berners-Lee predicted a Semantic Web (Ontologies, Large Scale Knowledge Graphs, ...) where images and text were explicitly linked via rigid, human-curated RDF triples and MPEG-7 schemas.

Today, a model like CLIP simply maps an image of a cat and the text "two cats on a couch" to nearby points in a shared continuous vector space using a contrastive InfoNCE loss.

If a simple matrix dot-product can ground visual reality to linguistic semantics better than decades of expert-curated symbolic graphs, was the entire symbolic AI paradigm fundamentally flawed, or are continuous embeddings just a highly compressed, uninterpretable version of the exact same knowledge graphs?

When a model simply maximizes cosine similarity, does it actually "understand" the concept of a cat, or is it just exploiting a statistical shortcut?



On the Synthetic Data Paradox: Why Doesn't Multimodality Collapse?

In earlier discussions, we established that training language models recursively on AI-generated text leads to 'model collapse'—an inescapable degradation of the data distribution.

Yet, the lecture reveals that state-of-the-art vision models like DALL-E 3 are trained on a mixture containing up to 95% synthetic, AI-generated captions (such as Descriptive Synthetic Captions from GPT-4V) to replace poor-quality human alt-text.

Why does synthetic data act as a fatal poison for pure text models, but serve as an essential performance-booster for multimodal text-to-image models?

Does injecting a second physical modality (vision) anchor the synthetic text to reality, fundamentally breaking the entropy loop that causes model collapse?

On the Universal Transformer: Is Everything Just a Sequence? Can MHSA extract all kinds of contexts from data?

We originally defined Transformers as language models designed to process 1D sequences of text tokens.

But in this lecture, we see the exact same architecture predicting the 3D folding of amino acid chains in AlphaFold, generating continuous robotic arm trajectories in VLA models, and performing 'Chain-of-Frames' visual planning in video models like Veo 3.

If the exact same self-attention mathematical operation can fold a protein, steer an industrial robot, and write a poem, does this imply that physics, biology, and human language share an underlying, universal sequential grammar?

Or are we simply forcing fundamentally different physical realities into the only sequence-prediction hammer we currently possess because it scales well on GPUs?